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Characterization and Modeling of Residential Electricity Demand for Flexible Load
Integration in Microgrids: The Case of Lebanon

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Abstract

Reliable electricity access still constitutes a challenge in many parts of the world. This is especially true in countries with weak energy infrastructure. Lebanon also faces similar problems. Indeed, the country still experiences frequent outages and a shortage of public energy supply due to its continuous electricity blackouts. Therefore, new decentralized energy solutions such as microgrids, diesel generators, and solar panels have emerged in the country. They aim to meet the energy needs of the population and ensure energy security. They can also integrate renewable energy into local energy mixes. Microgrids have therefore become a practical way to increase energy access. Indeed, they can store excess energy for times of low production and balance the supply and demand of power. Mostly, microgrids operate independently. However, they can occasionally connect to the main grid as a backup, with battery storage and intelligent management systems. Microgrids require demand management, cost optimization, and energy planning for operational and financial purposes. In Lebanon, microgrids have become popular in villages as an alternative to traditional energy sources. It is therefore interesting to study the case of Bechaaleh, a village in Lebanon, located at around 1,200 meters of altitude. The microgrid of Bechaaleh has 1.6 MWh of battery storage and 300 kWp of solar capacity. It showcases how decentralized systems can increase the proportion of renewable energy in the local energy mix while ensuring inhabitants have access to electricity. The objective of the thesis is to model electricity demand in Lebanon and validate the energy model with the Bechaaleh data. To increase the adoption of renewable energy while improving energy security and affordability, this report aims to offer a replicable framework for sustainable microgrid deployment in Lebanon and comparable contexts through demand modelling.

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1 Introduction

The urgent need to improve energy security, decarbonize electrical systems, and provide cheap access to dependable power is causing a significant shift in the global energy landscape. Therefore, microgrids have become a viable option since they provide flexible, resilient, and localized energy systems that may integrate renewable energy sources while being stable and reasonably priced. Microgrids solve many of the drawbacks of conventional centralized power systems. In fact, they decentralize electricity generation and integrate control mechanisms, especially in areas with rising energy demand and frequent supply disruptions.

Given its frequent electricity shortages, significant reliance on expensive fossil fuels, and fragmented power delivery infrastructure, Lebanon makes a particularly strong case for the research and implementation of microgrids. To design microgrids, it is important to understand household electricity demand and its variance at the appliance level, demographics, and climate zones. This thesis offers a framework to properly model electricity demand at the appliance level while taking demographics, seasonal fluctuations, and variable loads into consideration. The model can contribute to larger discussions on energy transition by offering insights into how microgrids use may be suited for Lebanon's socioeconomic and climatic characteristics. While microgrids offer reliability and serve as a solution to the lack of electricity provided by the government, they also require energy planning and cost optimization. Bechaaleh, a small village in Batroun, Lebanon, at around 1200 meters in altitude, has adopted a microgrid as its main electricity source. Its microgrid has a 300 kWp solar capacity and 1.6 MWh battery storage. Through the microgrid, energy is harnessed using renewable energy such as solar and wind sources. Energy supply surplus is stored in advanced battery systems for use during low-production periods. This ensures energy supply with smart grid management tools and optional grid connectivity for backup. The village of Bechaaleh is therefore sustainable, given that the microgrid has allowed it to increase its share of renewables in its energy mix. This creates a case for the analysis of the demand for electricity within the village. The methodology integrates mainly time-series analysis and energy modelling.

The study can be used as a basis for creating a dynamic tariff design. By validating the model on Bechaaleh's dataset, the aim is to offer a replicable framework for other villages facing similar energy challenges in Lebanon.



Figure 1: Microgrid of Bechaaleh. Source: <https://megreen.energy>

2 Literature review - Microgrids

2.1 Introduction

According to the United States Department of Energy, a microgrid is “a group of interconnected loads and distributed energy resources within clearly defined electrical boundaries that acts as a single controllable entity concerning the grid. A microgrid can connect and disconnect from the grid to enable it to operate in both grid-connected or island-mode” (Ton & Smith, 2012). Research on the microgrid began in 1999 when the Consortium for Electric Reliability Technology Solutions (CERTS) first recognized today's grid-connected microgrid concept (Hirsch, Parag, & Guerrero, 2018). To further explain, the microgrid constitutes a localized system composed of distributed energy resources, including solar panels, wind turbines, fuel cells, energy storage, controllable loads, and intelligent control systems. In their design, microgrids connect and disconnect to a wider grid, which allows them to supply local loads during outages or disturbances on the main network (Ton & Smith, 2012).

2.2 Classification

Microgrids can be classified according to different criteria. To start with, microgrids are classified based on their grid connectivity. Microgrids are either grid-connected, operating in parallel to the utility, while stand-alone or off-grid microgrids operate in isolation. Another classification is the electric configuration of the microgrid:

- AC microgrids are built around conventional alternating-current distribution infrastructure.
- DC microgrids use direct-current architecture.
- Hybrid AC/DC microgrids combine both and are increasingly explored for efficiency in integrating DC sources like photovoltaics (Justo, Mwasilu, Lee, & Jung, 2013).

Microgrids are classified based on scale and application. For example, remote community microgrids, campus or institutional microgrids, are different than nano grids, which are small-scale systems or larger multi-megawatt systems (Uddin, et al., 2023). Moreover, there is a distinction that can be noted between customer microgrids, which serve a single facility or campus, and utility or community microgrids that can span multiple customers or an entire community (Chandak & Rout, 2020).

Regardless of their classification, microgrids share the common feature of operating as a coherent system that can self-supply and manage local energy needs. In recent literature, the microgrid concept and its classifications have been reviewed comprehensively, reflecting new updates in definition since its initial formulation (Uddin, et al., 2023).

2.3 Technical Components and Architecture

2.3.1 Technical components

The microgrid consists of many components that are integrated into a local network. Distributed generation (DG) constitutes the primary components of the electricity generation system, which can include renewable sources such as solar photovoltaic and

wind turbines, and small generators (diesel gensets or microturbines) (Cagnano, 2020). Hybrid systems combining solar panels with diesel backup integrate multiple technologies. The aim of this integration is to enhance reliability and ensure continuous power for households. Energy storage constitutes a key energy storage component. Batteries are commonly used for energy storage, providing energy buffering, frequency regulation, and backup power. However, other options are available for energy storage such as flywheels or supercapacitors, which are used for power quality support.

A microgrid includes residential, commercial, or industrial loads that are designated as critical or non-critical during islanded operation. These are tied together through a low-voltage distribution network that forms the physical grid of the microgrid (Planas, Gil-De-Muro, Andreu, Kortabarria, & De Alegría, 2012).

A controlled point of common coupling (PCC) connects the microgrid to the utility grid, usually through a circuit breaker that can isolate the microgrid based on needs (Chandak & Rout, 2020). Finally, intelligent microgrid control systems can oversee the coordination of generation, storage, and loads, which allows the maintenance of stable operations by ensuring a balance between supply and demand. Intelligent microgrid control systems play an important role in ensuring the reliable and stable operation of distributed energy resources. Intelligent microgrid control systems are essential for maintaining stability due to the intermittency of renewable sources and the variability of demand. Often, a hierarchical control framework is adopted (Ahmad, 2023).

These control methods use multi-agent systems, in which each distributed energy resource, energy storage unit, or controllable load is represented by an autonomous software agent. These agents communicate and cooperate to achieve local objectives, such as optimal power sharing or state-of-charge management, while contributing to global goals such as cost minimization or emission reduction. This distributed control approach improves scalability, resiliency, and flexibility in both grid-connected and islanded operating modes (Logenthiran, Naayagi, Woo, Phan, & Abidi, 2015). Real-time data supports the coordination capability of these systems. They can indeed respond within milliseconds to changes in network conditions. This reduces power delivery disruptions and enhances system efficiency. Intelligent microgrid control systems are therefore

important to integrate renewable energy sources while preserving operational stability and financial performance.

2.3.2 Architecture and control levels

Hierarchical control can be used to understand the architecture of the microgrid. Primary control uses inverter-based droop control and stabilizes voltage and frequency locally at the field level. Secondary control operates on a management level. This level of control corrects frequency and voltage deviations and balances load sharing among distributed energy sources (DERs). At the tertiary control level, a synchronization loop transfers the microgrid from islanding to grid-connected modes. Nowadays, most microgrids are AC-based as they can utilize conventional AC equipment and interface with the legacy grid. However, DC or hybrid AC/DC microgrids have gained popularity since they are used in maritime applications, datacenters, and telecom systems. The overall architecture of the microgrid is defined by its flexibility and modularity. Each specific context can have tailored components and configurations such as rural areas incorporating renewable energy as an add-on to diesel generator-based systems. The hierarchical control structure ensures that components of the microgrid operate cohesively as a single self-governing grid (Hirsch, Parag, & Guerrero, 2018).

2.4 Key benefits of microgrids

2.4.1 Reliability and Resilience

Given the wide range of benefits that microgrid offers, interest in the technology is rising. Microgrids offer a wide range of benefits, such as the improvement in reliability and resilience of power supply of critical facilities (transportation, communications, waste treatment, healthcare, food, and emergency response infrastructure).

This resilience can be explained by the fact that microgrids can continue providing electricity to critical facilities or communities during wider grid outages or emergencies (Hirsch, Parag, & Guerrero, 2018). Indeed, microgrids have operational independence

during grid failure due to the “islanding” capacities, which allow the microgrid to operate if the main grid fails. Indeed, literature states that countries that are more likely to face natural disasters are also more likely to adopt microgrids as an energy provider (e.g. Japan’s integration of microgrids after the 2011 earthquake). One study conducted a fixed effects panel data regression using data from U.S. states between 2011 and 2017 to identify the main factors influencing microgrid adoption. One of the main hypotheses of the study was that states experiencing more frequent natural disasters are more likely to adopt microgrids to increase resilience. The regression model includes a variable for disaster frequency that measures the frequency at which a state experiences natural disasters. The findings indicate that disaster frequency statistically predicts the number of microgrid projects (Ajaz, 2018).

2.4.2 Integration of renewable and clean energy sources

Another important advance of the microgrid is its integration of renewable and clean energy sources creating environmental and economic benefits. Microgrids provide a platform to integrate distributed renewable energy such as solar PV or wind energy. Indeed, these energy sources are difficult to integrate into a centralized grid. Microgrids can therefore reduce intermittence by increasing energy storage and managing the fluctuations in renewable energy output. This enhances the reliability of renewable energy (Akorede, Hizam, & Pouresmaeil, 2010)

Moreover, the implementation of microgrids increases energy efficiency and creates cost savings through strategies such as peak shaving. Peak shaving is an energy management strategy used to reduce electricity demand during peak hours. It is particularly useful when energy prices and grid stress are high. Peak shaving reduces energy consumption by decreasing the energy used from the grid during peak hours. It also optimizes the use of distributed energy resources. Therefore, peak shaving reduces stress on the main grid and creates a sustainable schedule for the use of battery or renewable energy (Uddin, et al., 2023).

2.4.3 Reduction in Costs

Many studies have demonstrated the fact that microgrids create savings through the reduction of energy costs given the implementation of renewable energy and the reduction of the need for fossil fuel (Shahzad, et al., 2023). By reducing the stress on transmission and distribution systems, implementing a microgrid can also defer or avoid expensive investments in grid infrastructure. Moreover, microgrids also reduce distribution losses and improve overall efficiency through the increase of on-site generation. This successfully allows utilities to manage demand more effectively without a substantial grid. Moreover, microgrids also empower communities or facilities to have better control over their energy sources and usage. This empowerment creates an increase in self-sufficiency and a greater protection against volatile utility supply or prices (Soshinskaya, Crijns-Graus, Guerrero, & Vasquez, 2014). This local empowerment can be thought of as “energy democracy”. According to the Center for Social Inclusion (CSI) in the United States, “Energy democracy means that community residents are innovators, planners, and decision-makers on how to use and create energy that is local and renewable”. This allows to take decisions that are better aligned with priorities such as incorporating cleaner energy or ensuring power to essential services (Ajaz, 2018).

To summarize, the microgrid enhances resilience, supports sustainability, and cuts costs at a local level. Many of the advantages incurred such as resilience by implementing a microgrid cannot be captured by purely economic metrics, but they play a crucial role in increasing microgrid adoption (Guibentif & Vuille, 2021).

2.5 Global trends

Microgrids have evolved from experimental concepts to mainstream components of electricity systems worldwide. Indeed, there are many experimental projects in microgrid research due to technological advancements, the growing interest in renewable energy, and the need for reliable electricity. In countries across Asia, North America, and Europe,

the deployment of microgrids is increasing to improve energy security, grid resilience and meet environmental goals.

This deployment results from a growing trend to integrate cleaner energy sources such as PV and fuel cells. The aim of the trend is to avoid relying solely on diesel generators (Lidula & Rajapakse, 2010). Nowadays, the majority of microgrids are found in North America and the Asia-Pacific (Uddin, et al., 2023).

In sub-Saharan countries, in 2013, around 70% of the population lived in rural areas, and around 10% had stable access to electricity. In such countries, microgrids with local renewable energy can constitute an alternative isolated system to promote socioeconomic development (Justo, Mwasilu, Lee, & Jung, 2013)

In Sweden, the Smart Grid Gotland shows how wind generation can be integrated on remote islands. In Germany, several demonstrations of microgrids have emerged as part of the country's energy transition (Soshinskaya, Crijns-Graus, Guerrero, & Vasquez, 2014).

Through active engagement and federal research programs, the United States have emerged as a leader in microgrid innovation.

In the United States, the adoption of microgrids by public power utilities showcases the increasing interest in emerging technologies. This adoption can be seen mainly in larger public utility companies since they have more resources and capacity to experiment. The main reasons for deploying microgrids are usually related to resilience, reliability and distribution infrastructure, especially in service delivery industries.

The four rationales for adapting microgrids are

- grid resilience and the continuity of service,
- showcasing clean energy and innovation,
- enhancing customer engagement, and
- operational optimization through islanding and distributed control.

However, despite the growing interest in microgrids, the implementation of microgrids remains limited across certain sectors due to regulatory requirements and resistance to change. Indeed, in the United States, the implementation of microgrids often remains in the pilot stage (Lenhart & Araújo, 2021).

2.6 Challenges in implementation

Although microgrids have many advantages, they still face numerous challenges that can prevent their implementation and adoption. These challenges are technical, economic, regulatory, and organizational in nature (Soshinskaya, Crijns-Graus, Guerrero, & Vasquez, 2014).

2.6.1 Technical challenges

Operating a microgrid is complex, especially in island mode. In fact, microgrids constitute complex systems given their range of distributed energy resources, requiring management and coordination to ensure the reliability of microgrid operations. In fact, maintaining voltage and frequency stability becomes significantly more difficult without the inertia of a large, centralized grid. Given that the integration of renewable energy requires forecasting, energy management systems and dynamic balancing mechanisms, operating the microgrid becomes more complex. Other challenges include issues during start-up of island mode or finding the best parameters through fine-tuning for energy management. Furthermore, another potential challenge in the implementation of microgrids is finding the appropriate design as poor design reduces the lifetime of microgrids (Uddin, et al., 2023). Traditional overcurrent can also be inadequate in microgrids, as the fault currents are usually lower, and power flows are bidirectional. This leads to the incorporation of more adaptive protection strategies for microgrids (Soshinskaya, Crijns-Graus, Guerrero, & Vasquez, 2014). Additionally, ensuring compatibility among diverse distributed energy resources further complicates the process.

Another key issue is cybersecurity. Indeed, as the grid relies on more information and communications technologies, its vulnerability to cybersecurity attacks increases. These

threats require a strong investment in cybersecurity, which is not the case for many microgrid projects (Hirsch, Parag, & Guerrero, 2018)

2.6.2 Economic challenges

On the economic side, the high upfront investment cost of microgrids remains an important barrier to entry. Although technical and regulatory barriers can be tackled, the commercialization of microgrids strongly depends on the ability to reduce the production costs of renewable energy generation, energy storage, and energy management systems.

Moreover, the complexity of the implementation of hybrid models might create an increase in electricity production costs. One of the main challenges of microgrids is that customers who generate their own electricity through net metering can still use the grid while buying less power from the utility. Because utility companies recover their fixed infrastructure costs mainly through electricity sales, customers who produce part of their own electricity through solar panels or other self-generation systems end up buying less power from the grid. As a result, they contribute less to covering these fixed grid maintenance costs. As more people start adopting these types of systems, remaining customers will start contributing more to grid costs. Additionally, the installation of microgrids in rural areas creates difficulties in maintenance and increases the costs of transportation.

Regarding regulatory challenges, since the microgrid industry is relatively new, standards for microsource integration, as well as safety and protection protocols, must be reviewed (Uddin, et al., 2023).

Finally, interconnection of the microgrid is crucial as microgrids that cannot connect to the utility grid must operate permanently in islanded mode, losing the ability to offer grid services and undermining their financial viability (Hirsch, Parag, & Guerrero, 2018).

2.6.3 Organizational challenges

Organizational and stakeholder barriers are considered a significant challenge in implementing microgrids. Some of the stakeholder-related issues include “conflicting self-interest and trust” and the lack of operational expertise among involved actors. (Soshinskaya, Crijns-Graus, Guerrero, & Vasquez, 2014). Given that the implementation of microgrids requires coordination amongst different entities such as utilities, local governments, and consumers, it can become complicated by uncertainty over ownership and management responsibilities. (Hirsch, Parag, & Guerrero, 2018)

Most of the microgrid projects were still in the pilot stage despite growing interest and technology maturity. This can be due to the absence of public engagement, which slows the deployment of microgrids. Additionally, training employees on energy management, control, and monitoring systems is crucial to support microgrid operations. However, there is often a lack of capacity, mostly in remote and rural areas (Uddin, et al., 2023).

2.7 Microgrids in Lebanon

Lebanon is an interesting case for the potential adoption of microgrids given its ongoing electricity crisis. Electricity du Liban (EDL) has been unable to supply stable electricity for decades, given the underinvestment, outdated infrastructure, and political problems the country faces. Lebanon’s situation created a need for alternative solutions to EDL such as diesel-based generators. These alternative solutions have become the main source of energy across the country.

Indeed, by 2018, diesel generators supplied 84% of households. These generators range from informal neighborhood solutions to commercial mini grids in local economies and political networks. Although diesel generators can compensate for electricity shortage created by the Lebanese state, they remain illegal under the current regulations. They are mainly costly and polluting. Generators currently serve the interests of political networks and importers of fuel. Given the monopoly and lack of transparency in fuel market, these actors continue to oppose reforms (Chaplain, et al., 2023).

In recent years, Lebanon has faced a severe economic crisis and fuel shortages. The collapse of the national utility has showcased the strong political power of diesel-importers and diesel-generators, as well as their inability to ensure energy security at an affordable cost. However, even at higher costs, decentralized renewable energy such as solar and hybrid mini grids have grown during the economic crisis. This transition serves as an addition to diesel generators rather than a full-scale replacement and has resulted in hybrid setups where photovoltaic (PV) panels operate alongside diesel generators and grid connections.

Several entities such as the UNDP and the Lebanese Center for Energy Conservation (LCEC) have promoted decentralized renewables since 2010. Financing initiatives such as The National Energy Efficiency and Renewable Energy Action (NEERA) financed 54% of 2160 installed projects between 2010 and 2019 (Chaplain, HAL – SciencesPo, 2022). In fact, the National Energy Efficiency and Renewable Energy Action (NEEREA) is a national financing mechanism that allows private sector entities to get subsidized loans for any type of energy efficiency and renewable energy projects. NEEREA is active through all Lebanese commercial banks under the leadership and management of the Central Bank of Lebanon.

The microgrid technologies in Lebanon have mainly served industrial and commercial firms, as well as wealthy households. Given the 2019 economic crisis, access to these types of technologies remains limited and unequal due to high upfront costs. The emergence of hybrid mini grids showcases a technical and socio-political shift. Energy access in Lebanon is becoming more privatized, localized, and stratified (Chaplain, et al., 2023). In this context, hybrid microgrids can be considered as a technical adaptation to unreliable electricity supply from the government and as socio-political arrangements of governance (Chaplain, HAL – SciencesPo, 2022). These energy systems sometimes face challenges regarding legal recognition, grid integration, and standardization since they operate informally, usually through private initiatives or donors. The regulatory environment remains unstable as established actors and elite stakeholders actively seek to maintain their control. To conclude, hybridization has allowed electricity generation in the

absence of state reliability. However, it has also created social inequalities and prevented an inclusive and sustainable energy transition in the country (Chaplain, et al., 2023).

On December 14, 2023, the Lebanese parliament adopted Law No. 318/2023. This law adoption constitutes a historic step to address Lebanon's energy crisis and increase the use of renewable energy technology. The distributed Renewable Energy (DRE) law can be used by private citizens, companies, municipalities, and licensed producers to generate electricity from renewable sources up to 10 MW in capacity. They can also use Lebanon's infrastructure for peer-to-peer transactions and net-metering. This gives the Electricity Regulatory Authority (ERA) the authority to determine eligibility requirements, grid-connection protocols, pricing, and templates for power purchase agreements. This law constitutes a major step towards reaching around 30% of renewable electricity in the country's energy mix by 2030. It also gives local communities and the private sector the ability to handle urban and rural electricity demands, which lowers reliance on expensive, polluting diesel generators. Although operationalizing the ERA and strengthening grid management constitute essential steps for full implementation, its approval marks a rare moment of agreement among lawmakers and an important move to attract private investment.

3 Methodology

This research relies on fundamental electricity market concepts such as demand curves, peak demand periods, and renewable energy challenges to model demand. The main steps of this research are the following: Data collection, demand modelling, and validation of the model with real-world data.

3.1 Electricity demand and supply fundamentals

To model electricity supply and demand curves, it is important to understand fundamental concepts in electricity markets. In power systems, supply must be balanced to meet demand. The demand curve represents the electricity consumption varying over time, while the supply curve represents the available electricity generation and its cost. Key

concepts shall be defined at the beginning of the methodology. Peak demand represents the hours at which the electricity demand is at its highest, usually in the evenings when households are at home. Since operating these power plants is expensive, it increases the cost of electricity during peak hours (Faruqui & Palmer, 2011). Moreover, there are two main types of loads: base loads and peak loads. Base load represents the demand that should be available constantly through cheaper generators. Peak load constitutes the excess of demand that is above the base load. The supply curve ranks available power sources by their marginal costs of production, from lowest to highest and structures them according to merit order. When demand is low, only the cheapest power sources operate. As demand increases, the market price for electricity increases. Finally, the duck curve must be explained. As solar power takes an increasingly significant part of overall energy consumption, the net demand curve can start to take the form of the “duck”. Indeed, as illustrated in the figure 2 below of California’s duck curve, in the morning (6-8 AM), demand increases but solar output is still low, which raises the net load. From 10 AM to 2 PM, solar power peaks which reduces the net load. Between 5 and 8 PM, Solar production drops while demand remains high, causing an increase in net load. Moreover, deeper midday dips can be seen over time (net load was 13–14 GW in 2015 and dropped as low as 4-5 GW in 2023). This indicates that solar energy is slowly replacing conventional energy sources.

California's duck curve is getting deeper

CAISO lowest net load day each spring (March–May, 2015–2023), gigawatts

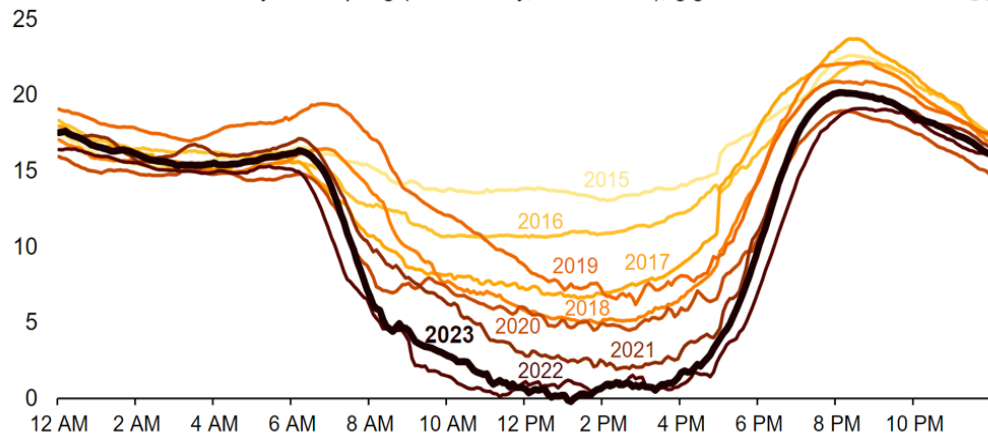


Figure 2: Net load curve illustrating the "duck curve" challenge. Adapted from Confronting the Duck Curve: How to Address Over-Generation of Solar Energy (n.d.-b), Energy.gov, <https://www.energy.gov/eere/articles/confronting-duck-curve-how-address-over>

In today's electricity market, managing demand-side management is becoming more crucial. Demand response (DR) represents incentives that push the end-users to consume electricity based on different supply conditions. In the case of microgrids, the ultimate objective of such incentive programs would be to push end-users to consume electricity at peak levels or when there is an excess production of electricity to reduce the peak of the duck curve. This can be achieved with dynamic pricing, with varying pricing for electricity throughout the day. Time-of-use pricing can be used as dynamic pricing to model the behavior of the customers. By offering lower prices during off-peak hours or high renewable output hours and higher prices during peak periods, the demand curve flattens as utilities can encourage load shifting (Bakare, Abdulkarim, Zeeshan, & Shuaibu, 2023).

These fundamental concepts are key to understanding the available data better. The case study of this thesis is a village in Lebanon where the aim is to characterize the supply of the village, as well as the village's demand profile to create a dynamic pricing strategy.

3.2 Electricity demand modelling approach

Demand modelling constitutes one of the core principles of this thesis. The objective is to create a demand profile for the village based on historical data. The first important concepts to be defined are bottom-up and top-down modeling. There are two demand modelling approaches. Top-down models treat the whole sector or area as an energy sink and often use econometric or aggregate time-series data, whereas bottom-up models build demand profiles from individual building or appliance data (Swan & Ugursal, 2009). These models will mainly tackle residential and community demand segmentation. Within residential demand, it is important to differentiate between primary or inflexible loads, such as lighting or refrigeration, which must be met when needed and flexible or deferrable loads such as running the laundry or certain appliances. For the modeling, primary demand should be met, whereas flexible demand can be met through price

incentives or signals. The suggested model will be validated with the Bechaaleh dataset and must be generalizable and scalable to other villages with similar sizes and energy sources. In the proposed modeling technique, given the nature of the available data, a time-series simulation model for demand will be used. The approach would be to derive an average 24-hour load curve, as well as a yearly consumption curve. Mostly, a simple representation of daily variation in consumption may be sufficient. Then, a baseline demand profile representing normal usage under the current flat tariff or practice can be generated. This profile will serve as the input for the next steps. The monthly consumption behavior of households will be studied to uncover new consumption trends.

A bottom-up modelling approach, which aggregates demand by modeling consumption at the end-user level, will be proposed. For instance, a bottom-up simulation of a multi-occupant household's demand can be analyzed by dividing total consumption into key end-use categories such as cold appliances, heating, ventilation and air conditioning (HVAC), lighting, and other activities driven by occupants (Muratori, Roberts, Sioshansi, Marano, & Rizzoni, 2013).

The modeling approach adopted in this thesis will start by representing households and their appliances before creating a village scale demand load. This method offers granularity as it allows to see how each appliance and each household contributes to the overall consumption of the village. This detail is crucial to characterize flexible loads and to understand which appliances lead to the peak consumption. In our model, based on the needs, we could use fine-tuning for appliance-level outputs to compare with other available metered data.

3.3 Flexible Load Characterization

A distinction between inflexible and flexible loads in the aggregated results will be made. In the Lebanese village context, inflexible demand refers to the critical loads that are priority and highly time dependent. These loads are the ones that users are unwilling or unable to shift in time. From the above categories, lighting, cooking appliances at mealtimes, and refrigeration would be classified as inflexible. These loads must be met as

they occur. Therefore, they form a baseline of demand that a microgrid must always be ready to supply. Flexible demand which covers loads that could be deferred or scheduled at different times with minimal impact on users includes the laundry, dishwashing, and other discretionary usage, as well as potentially certain heating/cooling adjustments. By characterizing each simulated appliance load as inflexible or flexible, the total inflexible load curve will be computed for the village and the total flexible portion. This is very useful for planning: for instance, the critical inflexible load curve represents the demand that must be met even under strict conditions. The flexible load curve represents the portion of demand that could participate in demand response or be moved during peak times. These will be used to examine scenarios such as: “How much peak reduction is possible if all flexible loads were moved off the peak hour?” The importance of this distinction is studied in prior research. Residential demand holds considerable flexibility potential, but its realization depends on human willingness to shift usage. The model outputs will explicitly show the magnitude of that potential flexibility for the case study village. For example, if the village evening peak is 100 kW and the model determines that 30 kW of that comes from flexible loads, including water heating and laundry, the peak could drop to 70 kW if those flexible loads were rescheduled. This kind of insight can guide the demand response programs (Williams, Gnoth, Hooper, & Chase, 2025).

3.4 Appliance level characterization and area dependency

The proposed model includes the segmentation of appliance consumption for households. This characterization could make the model more transparent, allowing to trace demand to different sources. The total household consumption can be divided into categories: cold appliances, space conditioning, lighting, and other usage driven by occupant activities (Muratori, Roberts, Sioshansi, Marano, & Rizzoni, 2013). This segmentation can be applied to different Lebanese villages. Below can be found the appliances. Appliance ownership rates for Lebanon were obtained from a survey conducted by Fadak Moussa (Moussa, 2020).

- **Lighting:** Lighting is tightly related to the size of the house and seasonal day patterns. Lighting load will be modelled per square meter and longer lighting hours in winter will be accounted for.
- **Heating, Ventilation, and Cooling (HVAC):** This category includes space cooling loads. In the case of Bechaaleh, at 1200m of altitude, heating in winter and cooling in summer might be present. HVAC usage depends on occupancy, which is the reason to include it separately as it can dominate peaks. HVAC systems can be modeled with simplified physical sub-models, such as an on-off control pattern based on a thermostat. This category is treated as area dependent. For example, a larger house with more rooms can have more ACs running concurrently, driving up peak demand.
- **Refrigeration (cold appliances):** Refrigerators and freezers run continuously with thermostat-driven cycles. Refrigeration constitutes the base load of the demand.
- **Domestic hot water:** Hot water energy use tends to scale with household size and house area. A larger home usually houses a bigger family, meaning more hot water consumption (more showers, etc.). Also, large homes might have multiple bathrooms or a larger tank. We treat Domestic hot water as semi area-dependent. All homes have at least one water heater, but a big household might use it more frequently (two heating cycles per day vs one for a small household)
- **Space heating:** Electric space heating is excluded in all scenarios, but note that if it were included, it would be highly area-dependent because larger volumes require more heating power. In the Lebanese scenarios, heating is non-electric in high mountain villages. In fact, electric heating is mostly used in the coastal zone where the heating demand in winter is relatively low.
- **Cooking and Kitchen Appliances:** Cooking-related demand typically peaks around mealtimes. These are generally present in most households (microwave ~64% ownership, electric oven ~48%). The number of microwave ovens or electric kettles do not scale with area. Thus, they are area-independent in count,

although total energy might be a bit higher in a large family. These loads are usually inflexible in timing, as people generally will not postpone cooking by hour.

- **Entertainment and Electronics:** This includes televisions, which are modelled separately, radios, computers, mobile phone charging, and general electronics. These are influenced by end-user behavior. These entertainment loads are generally inflexible in the short term as people usually watch devices at desired times.
- **Cleaning and Domestic Appliances:** This covers washing machines, dryers, dishwashers, vacuum cleaners, water pumps, and similar intermittent appliances. These appliances are deferrable. In this model, they will be considered as flexible demand and could treat their usage with schedule options.
- **Laundry Appliances:** Almost all households have a washing machine (~98%) irrespective of size. A larger household will do more loads of laundry per week, but the machine count remains one. Therefore, we treat the machine as area-independent (one per home), but we will increase the frequency of use for larger scenarios. This frequency scaling can be tied to occupancy count, which correlates with area in our scenarios. Dishwashers, as noted, are very rare (~6% ownership). Therefore, they are excluded by default from all scenarios.
- **Other:** These small appliances are generally present in most homes (vacuum ~82%, irons ~79% ownership) and do not depend on house size: a one-bedroom apartment can have a vacuum cleaner and an iron just as a big house does. The difference is that larger families may use them slightly more often (e.g. more frequent vacuuming). For modeling, the “other” category can be aggregated and assigned a modest daily energy of around 0.5-1 kWh/day.

3.5 Seasonal variations and temperature:

Seasonal variations are explicitly handled in the model to ensure the demand profiles reflect temporal patterns observed. The simulation will be divided into seasons: winter and summer and potentially low season if needed. Weather-sensitive loads like heating

and cooling are the most obvious. The HVAC takes seasonal temperature data as an input, so it generates higher loads on cold days or hot days as appropriate. Temperature can be incorporated into the analysis by creating a load profile for an average hot day and an average normal day (Habib & Hossain, 2025). Representative daily temperature profiles can be used for each season in Bechaaleh at 1200m; the village may have cold winters requiring heating and mild summers with some cooling. It is crucial to capture these trends, as confirmed by Literature. High-resolution models should accurately reflect seasonal variations in household load profiles realistically (Marszal-Pomianowska, Heiselberg, & Larsen, 2016) (Muratori, Roberts, Sioshansi, Marano, & Rizzoni, 2013). By calibrating the model across seasons and validating against any available seasonal consumption data from the village, these variations become better reflected. To sum up, by accounting for seasonality, the model's load curves become context-aware, which is crucial for microgrid planning.

3.6 Climate Zones in Lebanon

Lebanon is typically divided into four climatic zones which influence how residents use certain appliances, especially air conditioners (AC) for cooling and water heaters for domestic hot water (DHW). We exclude electric space heating from our electric load modeling, since most Lebanese households use non-electric heating sources (diesel, gas, or wood) in winter except for individual apartments in the coastal zone. For example, a high-mountain rural home may have significant heating needs in winter, but those are met by a wood stove or diesel furnace. Therefore, the electric profile in winter remains low aside from lighting and appliances. The focus will be on cooling, lighting, appliances, and water heating, which are included in the electric demand.

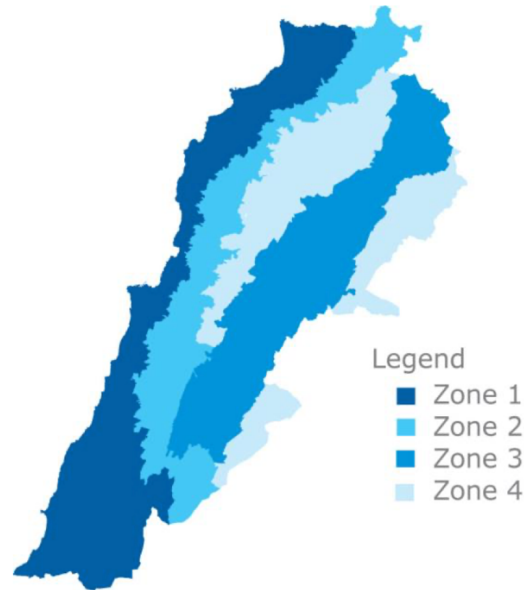


Figure 3: Climate zones in Lebanon- Source: Project GEF LEB/99/G35 au Liban

Zone 1 - Coastal (Beirut and coastal cities): Warm humid summers, mild winters. Households in Zone 1 almost universally have at least one air conditioner, used extensively during the hot summer months. Even small city apartments typically have a single AC unit often in a bedroom or main room that is used on very hot days or at night for sleeping. Larger homes may have multiple AC units to cool more space. AC usage in this coastal zone is heavy in summer, often every evening and night when temperatures and humidity remain high, and sometimes during midday if occupants are home. In winter, the temperatures are moderate, as loads consist mainly to lighting, hot water, and appliances. Electric water heaters are common, but usually used once a day to warm water, and possibly twice a day depending on weather. Lighting usage has shorter usage in summer due to late sunset compared to winter.

Zone 2 - Western Mid-Mountain (medium altitude): Warm summers, cool winters at mid-elevations. In these areas such as 500-800m altitude towns, summers are slightly cooler and less humid than the coast. AC ownership is still common, but usage is more moderate. Given the altitude, some houses may only have one AC unit or rely on natural ventilation or fans for hot days. AC is usually only used in the hottest days of the year. Winters are colder here than in Zone 1, so people rely on stoves or fuel heaters. Electric

water heaters might be run for longer durations in winter mornings and possibly an extra hour in the evening for higher demand. Overall, electric consumption in Zone 2 peaks on summer days and winter evenings, but AC energy use is lower than in the hotter zones. Lighting needs are slightly higher in winter due to longer nights, similar to other zones.

Zone 3 - Inland Plateau (Continental such as Bekaa plain): Hot dry summers, cold winters. This zone sees extreme seasonal differences. Summers are very hot (often $>35^{\circ}\text{C}$), so air-conditioning is widely present and heavily used. In larger homes, 2 or more AC units may be installed to cover multiple rooms, and during peak heat waves they might run continuously through the day when the family is home. However, because the climate is dry, nights can cool off and households often turn AC off overnight if outside temperatures drop sufficiently. In winter, this zone is quite cold. Homes usually use diesel boilers, wood stoves, etc. for space heating. Therefore, winter electrical demand in Zone 3 is lower than summer, as it consists of lighting, appliances, and water heating only. Electric water heaters are heavily used in winter, whereas in summer a single daily cycle or shorter usage may be enough. Overall, Zone 3 households have the highest summer peak loads due to cooling among all zones and pronounced seasonal variation.

Zone 4 - High Mountain such as Cedars, Faraya: Mild summers, very cold winters. In high-altitude areas, summers are cool enough that many households have no air conditioners at all or have units that are used. Even if a large rural house owns an AC or two, in a cooler mountainous village the AC usage is minimal, people often rely on the natural cool air, using fans or open windows on the rare warm afternoon. Thus, summer electric demand in Zone 4 is relatively low, driven mostly by appliances, lighting and sometimes fans. Winters, however, are harshly cold with snow and sub-zero temperatures. As elsewhere, heating is supplied by non-electric means (fuel/wood heating is the norm), so we exclude electric space heating from the load profile. The absence of electric heating means that, despite the cold, the winter electric load in Zone 4 is comprised of lighting which is used extensively during long dark winter evenings, water heating, and other appliances. Notably, some high-mountain homes have diesel-based central heating systems that also provide domestic hot water, in which case an electric water heater might

not be present or used in winter. Zone 4 households have relatively flat electric profiles compared to Zone 3 with lower summer peaks with little or no AC usage and only modest winter evening peaks (lighting, hot water). In essence, a high- mountain household's peak electrical demand might occur on a winter evening when lights and water heaters are on simultaneously (but still much lower than a coastal or inland home's summertime peak with ACs running).

3.7 Appliance ownership and usage variations by zone

Aside from climate-driven usage of AC and water heating, most other appliances have a more uniform presence across zones. According to survey data from the report of Fadak Moussa, nearly all Lebanese households have certain basic appliances such as refrigerators, TVs. Washing machines are 97% present, while some appliances are far less common as only ~6% have dishwashers (Moussa, 2020). These low penetration devices can be excluded from a "typical" household model to simplify. In the model proposed, dishwashers will be omitted given its rarity. Cooking appliances such as electric ovens, microwaves, kettles may see slight usage differences by region, but climate does not heavily change their usage. Laundry appliances such as washers, dryers and others like vacuum cleaners are common (washer ownership ~98%, vacuum ~82%) in all zones, though rural households might line-dry clothes rather than use electric dryers (Moussa, 2020). One notable difference is lighting usage, which can be higher in larger rural houses with more rooms to light and in winter months with longer nights, whereas a small urban apartment uses fewer and smaller lights. Another difference is that some cold-zone households use alternative water heating (solar or fuel boilers), whereas warm-zone households more often use small electric water heaters. The proposed model here will account for this by enabling or disabling the electric heater per scenario. In summary, the four climate zones primarily affect cooling and heating-related appliance use (AC and DHW), while ownership of most other appliances is fairly consistent nationally with the exception of luxury appliances (Bakare, Abdulkarim, Zeeshan, & Shuaibu, 2023).

3.8 Demographic variability and household size:

Households differ in family size, income, appliance ownership, and habits, leading to significant variability in consumption. The proposed model is designed to showcase each household's unique characteristics. This introduces behavioral and demographic variability as an optional but recommended feature. Rather than modeling an average household and multiplying it, many diverse households are used to reflect a real village's heterogeneity. The number of occupants per household can vary. A larger family generally uses more electricity than a single-person household. Occupant count can be incorporated as a parameter. For instance, a village with a mix of household sizes (single, 4-person, etc.), possibly based on the village data or local statistics can be generated (Marszal-Pomianowska, Heiselberg, & Larsen, 2016). Based on this, load curves for each household size can be obtained. However, for our study, climate zones and dwelling size categories were incorporated.

Combining the climate and dwelling size factors, 12 representative household scenarios can be defined. Each scenario is a unique pairing of climate zone (1- 4) and dwelling size category (small ~ 80 m², medium ~ 120 m², large ~ 200 m²). These numbers were derived from Fadak Moussa's study (Moussa, 2020). By varying both climate and size, a wide range of usage patterns can be captured. For instance:

- Zone 1, 80 m²: Small urban coastal apartment, 1-2 occupants. One AC unit used on hot summer nights, one fridge, minimal lighting, and low overall usage.
- Zone 1, 200 m²: Large Coastal villa, more than 5 occupants. Multiple ACs running in summer, many lights and electronics, and high overall demand.
- Zone 4, 80 m²: Small high-mountain cottage, couple or small family. Perhaps no AC (cool climate), heating by wood stove, so very low winter electrical load; a modest fridge and appliance usage.
- Zone 4, 200 m²: Large Mountain home, extended family. Possibly 2 AC units installed but used due to cool weather: highlighting use on winter evenings, heavy water heating in winter, multiple fridges/TVs, but still lower summer peak than a coastal equivalent

Each scenario specifies which appliances are present and their usage profiles. If a certain appliance is unrealistic for a scenario, it will be adjusted accordingly. By generating a load profile for each of these 12 scenarios and then averaging or aggregating them, the demand of a diversified village or region can be modeled.

3.9 Model structure and replicability:

An important design principle is replicability, which ensures the clarity of the understanding of the model and the ease of adapting it to other contexts. The model consists of several components or sub-models, each of which represents one aspect of the demand modelling. These components can be modified or extended independently. This mirrors the approach in recent models. For instance, in an agent-based framework, each appliance use (such as water heating and space heating) has its own sub-models that are coordinated together with programs (Williams, Gnoth, Hooper, & Chase, 2025).

4 Results

4.1 Model assumptions and results

Table 1 represents the assumptions and parameters considered in the proposed model.

Table 1: Assumptions and parameters considered in the proposed demand model

Appliance	Power (kW)	Usage Schedule	Notes
Lighting	$0.01 \times \text{area}$	Nighttime hours (18:00-06:00); usage increases in winter and in larger homes	Scales with floor area and season
Refrigerator	0.15 (constant)	Runs 24/7; averaged to reflect compressor cycling	Present in all households
TV	0.10 (when on)	Evenings (varies by usage profile): 19:00-23:00 typical	Longer runtime in large households or weekends
Washing Machine	0.50 (per cycle)	Probabilistic, during daytime (08:00-20:00); more loads on weekends	Number of weekly loads varies by household type
Water Heater	1.50 (when active)	06:00-08:00 and/or 18:00-20:00; usage varies by climate and family size	Disabled in some cold-zone households (e.g. those with diesel boilers)
AC Unit	1.00 (per unit)	Summer only; runs when temperature $> 21^\circ\text{C}$ and household is occupied	Number of ACs scales with house size and zone; zero in some Zone 4 households
Others	0.02 base + 1.0 spikes	Random short bursts throughout the day (microwave, kettle, vacuum, laundry)	Adds realistic noise to profiles without needing to simulate every device individually

Table 2 shows the specifications used in the model based on the previously defined assumptions and parameters.

Table 2: Specifications used for the code

Appliance	Condition (Time/Season/Usage)	Power Used (kW)	Notes
Lighting	hour $\geq$ 18 or hour $<$ 6	$0.01 \times \text{area} \times \text{seasonal} \times \text{behavior}$	Nighttime only. Boosted in winter months (Dec–Feb). Behavior $>150\text{m}^2$.
	Else (daytime)	$0.01 \times \text{area} \times 0.01$	Minimal standby use
Refrigerator	Always	$0.15 \times \text{fridge_count}$	Continuous base load
TV	high: 8-24, medium: 18-23, low: 19-22	0.10	Depends on usage level and hour
AC	if summer (day 152-273) and temp $> 21^\circ\text{C}$ and home	$1.00 \times \text{ac_units}$	Based on sinusoidal temp curve and occupancy
Water Heater	high: 6-8 and 18-20	1.50	High usage = morning and evening use
	medium: 6-8 + winter evenings (Nov-Feb)	1.50	Medium usage = less frequent evening use
	low: only 6-8	1.50	Minimal usage
Washing Machine	8-20 if random $<$ probab_hour (higher on weekends)	0.50	Probabilistic: usage = 2, 4, or 7 loads/week based on profile
Others	Always	$0.02 \text{ base} + 1.00 \text{ spike (1\% chance)}$	Random small base + occasional rare spike load

Table 3 presents the assumed climate zone parameters

Table 3: Assumptions related to the four climate zones

Zone	Description	Summer High ($^\circ\text{C}$)	Winter High ($^\circ\text{C}$)	AC Usage	Water Heating
Zone 1	Coastal	32	18	High (1-3 units)	Electric heaters standard
Zone 2	Mid-Mountain	30	12	Moderate (1-2 units)	Electric heaters standard
Zone 3	Inland	36	10	High (2-3 units)	Electric heaters standard
Zone 4	High-Mountain	25	5	Rare (0-1 units)	Often fuel/boiler-based (electric disabled in model)

The model is implemented using the Python programming language. The simulation runs over the full year for 8760 hours, as well as for a typical day. Behavior schedules and usage profiles reflect the different conditions, such as time-of-day rules and climate-dependent triggers, which simulate occupancy and appliance patterns. In the Python code, each appliance is modelled separately and then aggregated to enhance scalability across multiple household scenarios. The results are plotted as graphs that reflect real-time consumption behavior.

Figure 4 shows the average daily power curve for 12 scenarios. Each is a different combination of area (1 to 4) and floor area (80m², 120m² or 200m²). The horizontal axis shows the hour of the day, while the vertical axis shows the average power needs in kilowatts (kW). Large floors on the ground in the same area have a higher demand for electricity. Some areas, such as zone 3, show higher values than others. In the early morning (0:00-5:00), demand remains stable at medium levels, with the largest consumption observed in larger buildings in areas 3 and 4. A decrease in demand can be observed between 6:00 and 9:00, but after that, electricity usage remains at the lowest level at the end and beginning of the morning (9:00 AM to 4:00 PM). At about 5 PM, demand rapidly increases, reaching the summit significantly by 6 PM. At 7 PM, Zone 3 at 200m² reaches approximately 5.2 kW. After this night's spike, the request settles at the steady load level. The steady load level is higher than the day, but lower than the peak, and continues until midnight. This model shows bimodal loads with a small midday dip and dominant night peaks under the influence of local characteristics and building size.

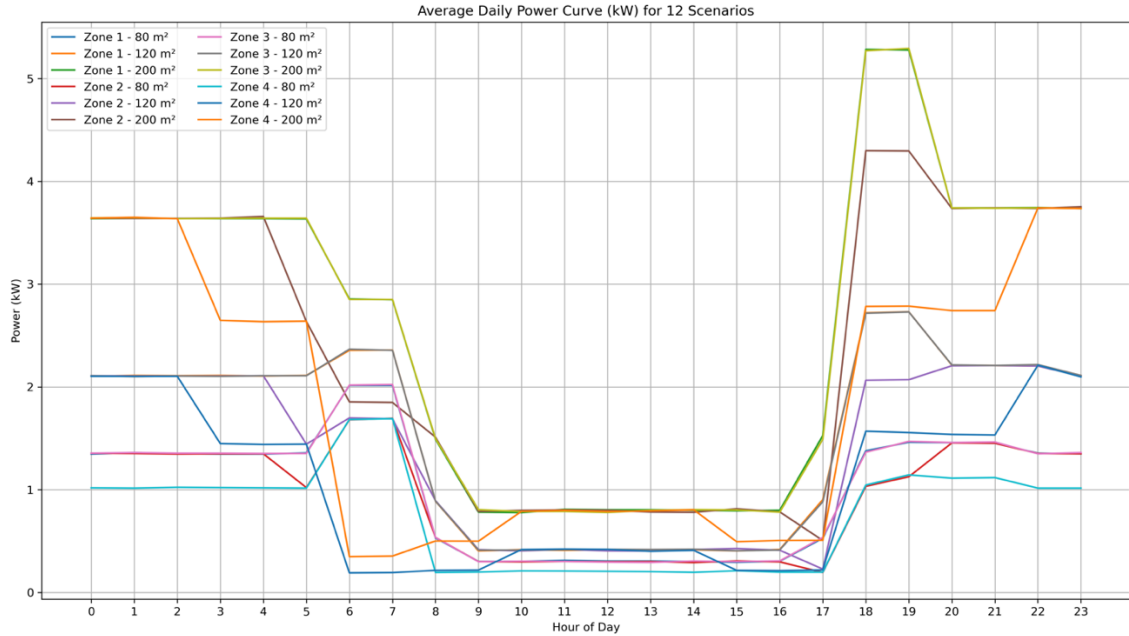


Figure 4: Average Daily Power Curve (kW) for 12 scenarios

Figure 5 shows the yearly load curves for 12 scenarios. They represent combinations of zone (1 to 4) and floor area (80 m², 120 m², or 200 m²). The x-axis plots the day of the year. The y-axis shows daily energy use in kilowatt hours (kWh). Each scenario showcases a clear seasonal pattern. Three operating phases can be observed. For phase 1, from day 1 to day 150, demand remains stable (phase 1). Larger floor areas consistently consume more energy than smaller ones, and Zone 3 scenarios shows the highest values among all zones. For phase 2, around day 150, a seasonal shift happens. This creates a clear periodic peak that lasts until about day 250. Large fluctuations in daily demand can be observed in the middle phase, as cooling or heating loads vary each day. Zone 3 at 200 m² reaches the highest peaks near 115 kWh per day. Finally, for phase 3, after day 250, consumption drops quickly, returning to levels seen at the start of the year.

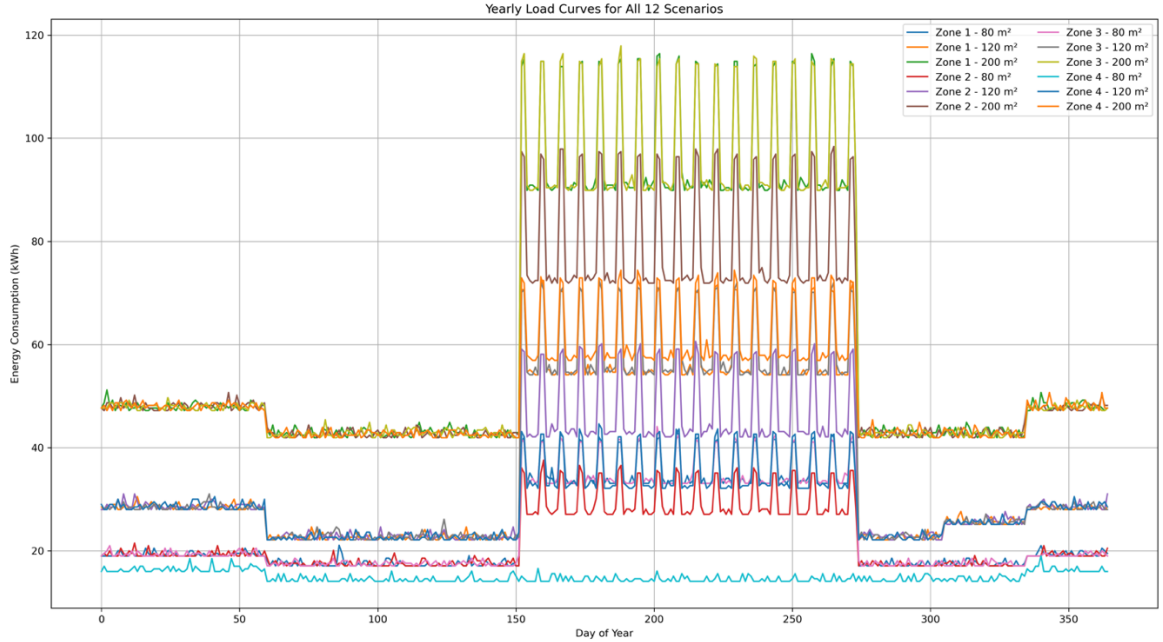


Figure 5: Yearly Load Curves (kWh) for all 12 scenarios

The mean daily load curve combines the 12 household scenarios according to the percentage distribution shown in Table 4. This estimates the average daily load (shown in Figure 6) of a Lebanese household combining all previously mentioned parameters (from the 12 scenarios).

Table 4: Assumptions related to the distribution of parameters

Climate Zone	80 m ²	120 m ²	200 m ²	Total per Zone
Zone 1 (Coastal)	10%	5%	5%	20%
Zone 2 (Mid-Mountain)	5%	10%	5%	20%
Zone 3 (Inland)	5%	10%	10%	25%
Zone 4 (High-Mountain)	5%	5%	5%	15%
Total per Area	25%	30%	25%	100%

An aggregation is conducted to reflect the composite daily demand of the modeled village. A general pattern remains similar to the individual scenarios with relatively high demand during the night and early morning hours before declining gradually between 6:00 and

9:00. Following a period of low consumption between mid-morning and late afternoon, demand increases sharply around 17:00 and reaches its highest values at 19:00. A slight decrease in demand follows this peak, as the load remains constant until midnight.

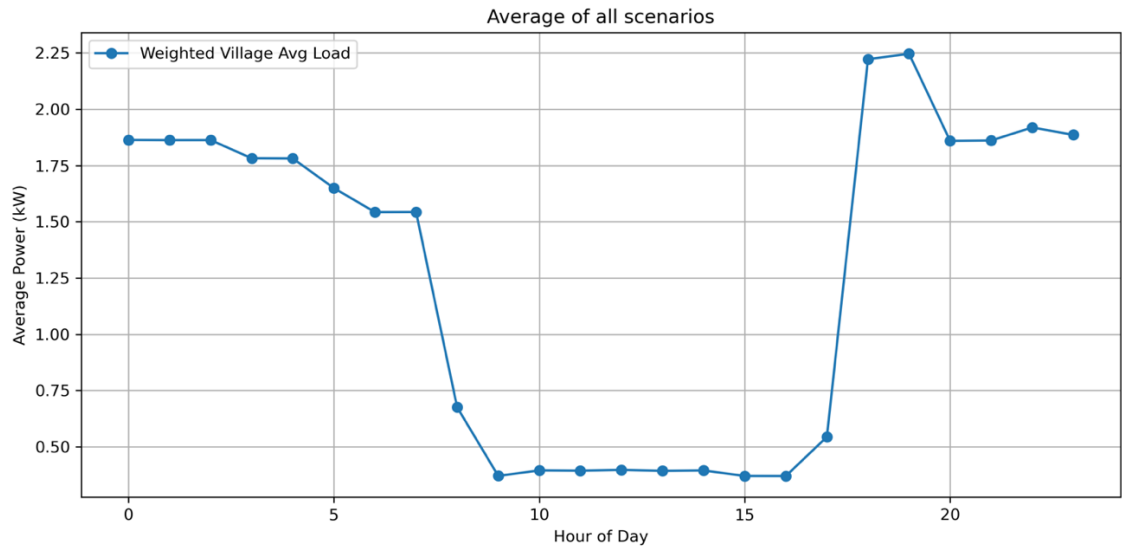


Figure 6: Average daily power profile (kW) across all scenarios

The average yearly consumption curve of a Lebanese household, shown in Figure 7, combines the 12 household scenarios according to the percentage distribution in a similar logic described above. The general pattern remains the same as the yearly load curves described above, with 3 well-defined phases.

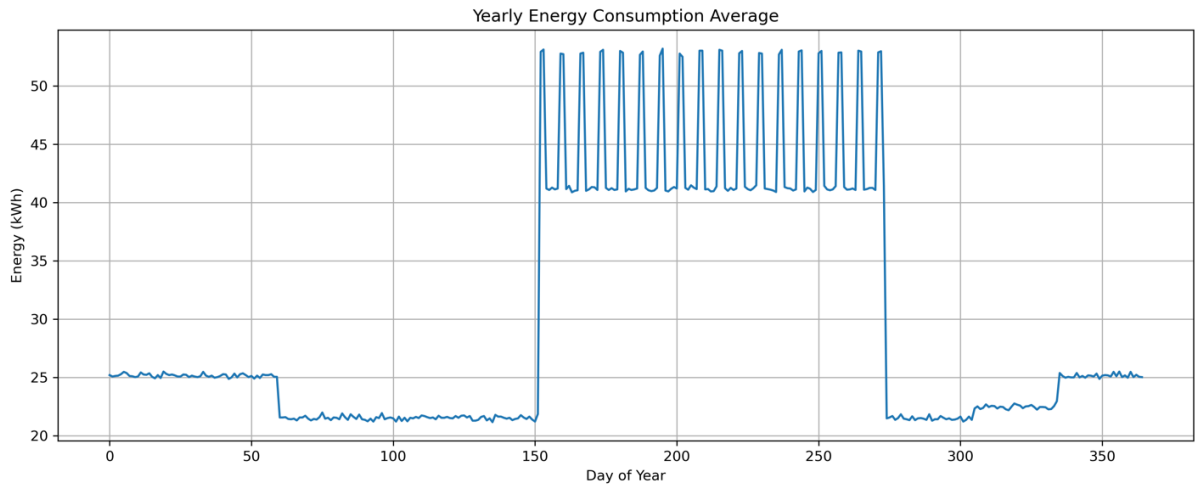


Figure 7: Average yearly energy profile (kWh) across all scenarios

Since Bechaaleh falls in zone 4, it is important to study consumption patterns for this zone separately. Yearly and daily load curves are also created for zone 4 separately. 3 surface areas were considered in a similar logic to the one discussed above. All surface areas were considered to have the same distribution as shown in Table 5.

Table 5: Assumptions related to the distribution of parameters (Zone 4)

Climate Zone	80 m ²	120 m ²	200 m ²	Total per Zone
Zone 4 (High-Mountain)	33.3%	33.3%	33.3%	100%

The yearly profile of Zone 4 (Figure 8) also shows a stable consumption during winter and spring, and a decline in consumption during summer. The overall shape of the chart differs from the average yearly energy consumption profile of Lebanese households, which typically shows higher consumption in summer. In contrast, Bechaaleh, located at the high altitude of Zone 4, displays a different pattern. Indeed, this is reflected by the absence of cooling during summer in zone 4, as well as more heating in winter. This trend is expected, given the absence of AC in zone 4 households. Moreover, a part of zone 4 households uses wood/biomass for heating, which diminishes consumption of electricity consumption in winter.

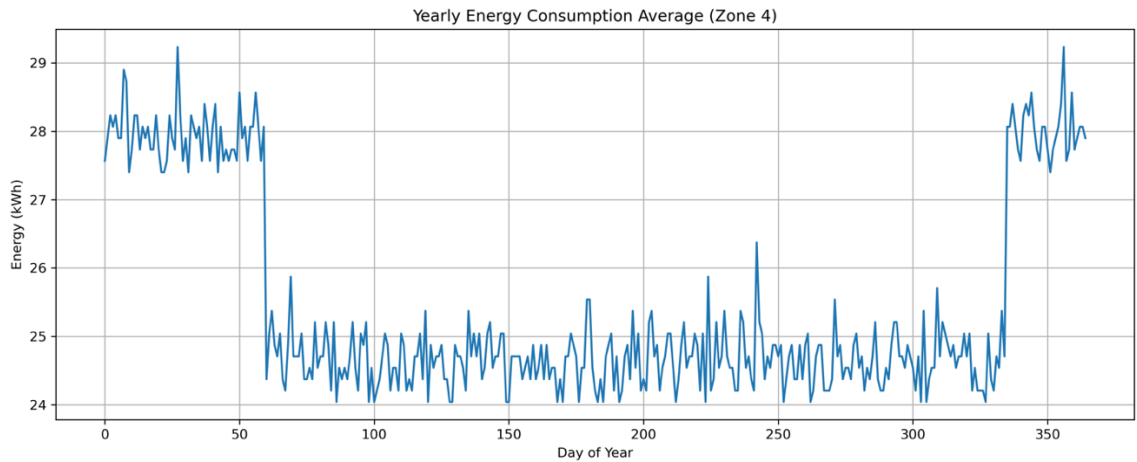


Figure 8: Average yearly power profile (kW) for Zone 4

The average daily curve for zone 4 (Figure 9) showcases a curve with high loads in mornings and evenings. A sharp drop in demand can be observed between 06:00 AM and

09:00 AM, with low consumption during the day. A second peak occurs between 6:00 PM and 8:00 PM. This peak can be explained by household occupancy, lighting, and water heating. During the day, demand remains low due to households being away and limited cooling in the mountains. This load profile highlights the strong influence of household routines and climate on daily consumption patterns.

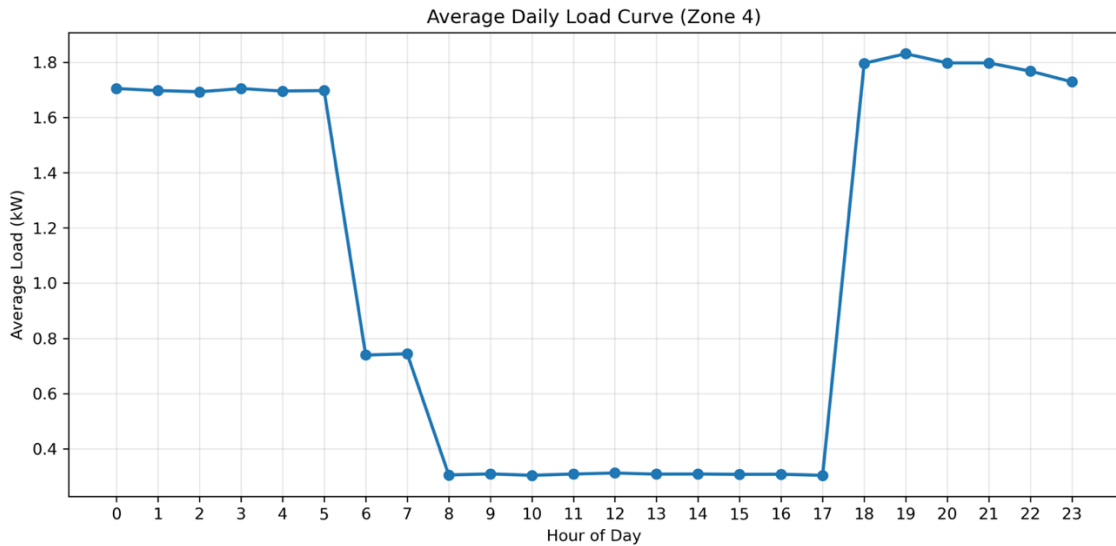


Figure 9: Average daily power profile (kW) for Zone 4

An analysis of the daily load curve is also conducted by season (Figure 10). The seasonal load curves displayed clear variations. In winter and spring, a peak can be noticed, which is linked to lighting and appliances. Overall, summer consumption is the highest, especially in the evening due to the use of AC at night. All seasons display a decrease in consumption during daytime while households are away from home.

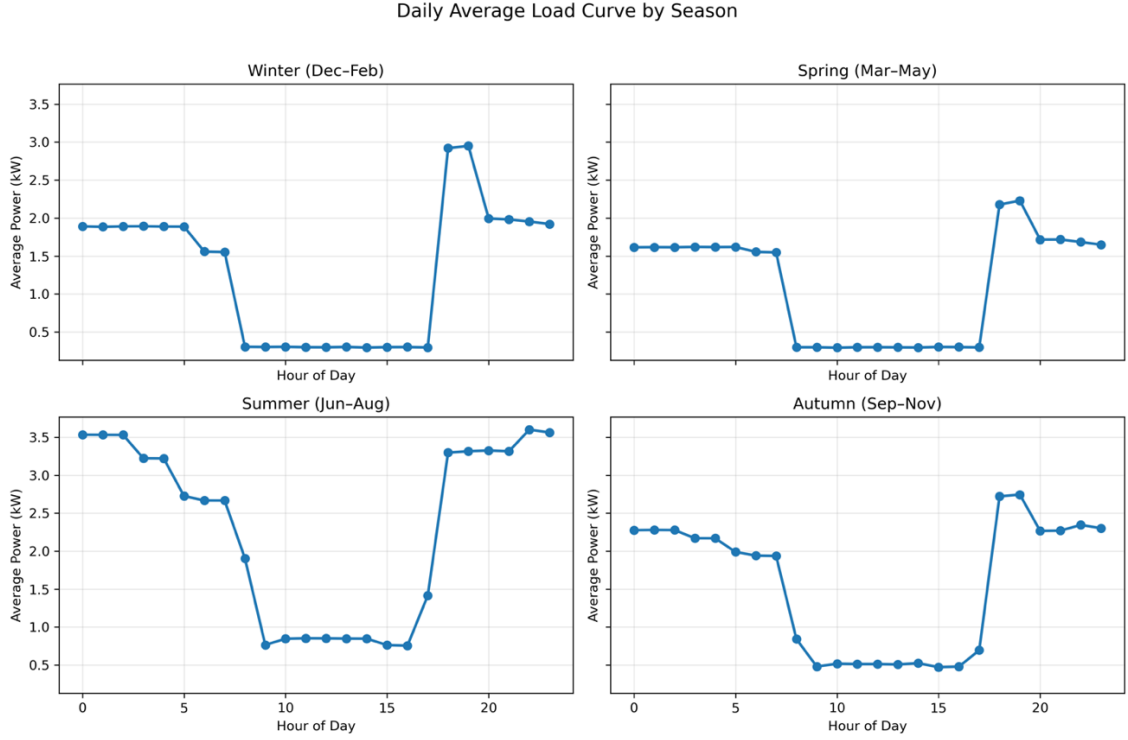


Figure 10: Average daily load profile (kW) per season

4.2 Data Collection

The modelling is applied to the village of Bechaaleh in the district of Batroun, Lebanon, for which data consumption per subscriber is available through [ME Green](#). The village's electricity consumption by residential consumption datasets includes time-stamped measurements of power consumption at the individual consumer level and at the distribution transformer (entire village). The data is available for the end of 2024 and the beginning of 2025. Through the dataset, insights on the following points can be obtained. The demand profiles constitute the data at the household level with timestamps at a couple of minutes intervals, which show the fluctuations of the consumption of end-users throughout periods of time.

The analysis of the electricity demand of the village of Bechaaleh serves a crucial role in the understanding of consumption patterns through the use of high-frequency data from the smart meters at household levels as well as at the transformer level.

4.2.1 Data understanding, preparation and cleaning

The raw data consists of CSV files collected from eight distribution transformers and approximately 150 residential smart meters. Each transformer has data gathered for two periods: before 2025 and after 2025. Data cleaning involved using the “concat” function in Python to obtain a centralized version of each dataset from the end of 2024 to mid-2025. Data cleaning also involved treating name inconsistencies, dropping missing values, and calculating the total instantaneous active power by adding the three-phase readings: ActivePowerP1, ActivePowerP2, and ActivePowerP3. The instantaneous power’s unit is in kW, which can be used for consumption modeling over time. Total active power at both individual and transformer level was calculated. Each transformer powers individual households, and the structure of the dataset can be represented with the example below in Figure 11.

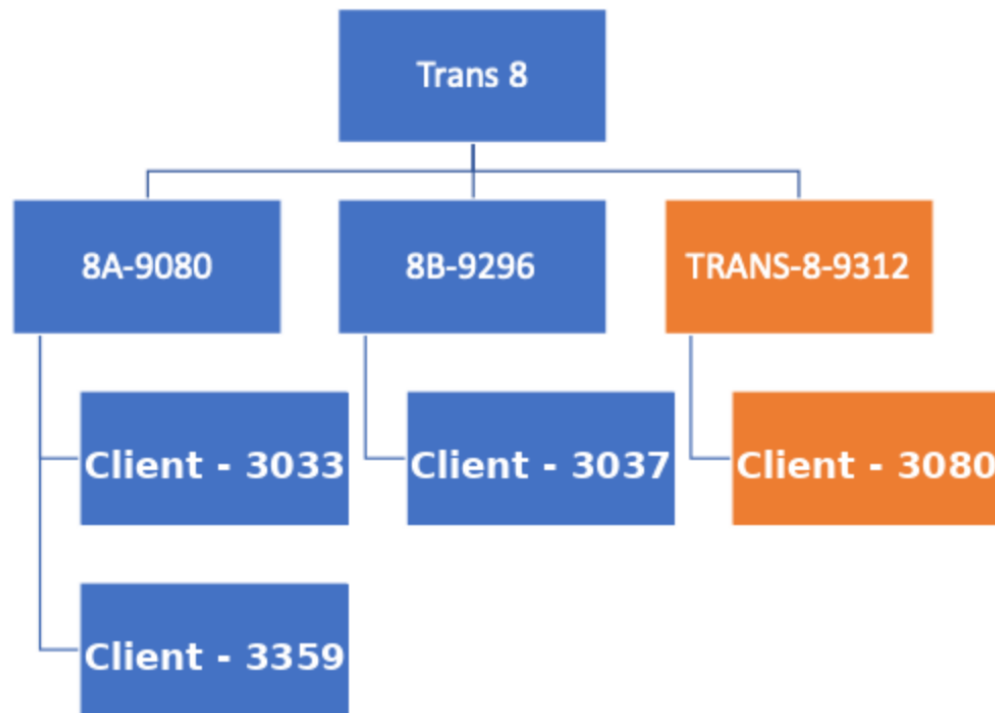


Figure 11: Example of the structure of transformer 8 in the dataset

Some individual households had a total instantaneous active power of 0, which were dropped from the individual analysis. Entries with valid timestamps and power measurements were retained to ensure the integrity of our data. At the transformer level, box plots were created to validate the reliability of the dataset. The results showed that the data falls within a realistic range. For most transformers, the median instantaneous power range falls between 1.0 and 2.5 kW, with upper range bars up to 6-8 kW at peak consumption. After noticing some disparities between total active power for transformer 1 and 3, the data structure was reviewed. While Transformer 1 supplies 37 households, Transformer 8 only supplies 4 households for example.

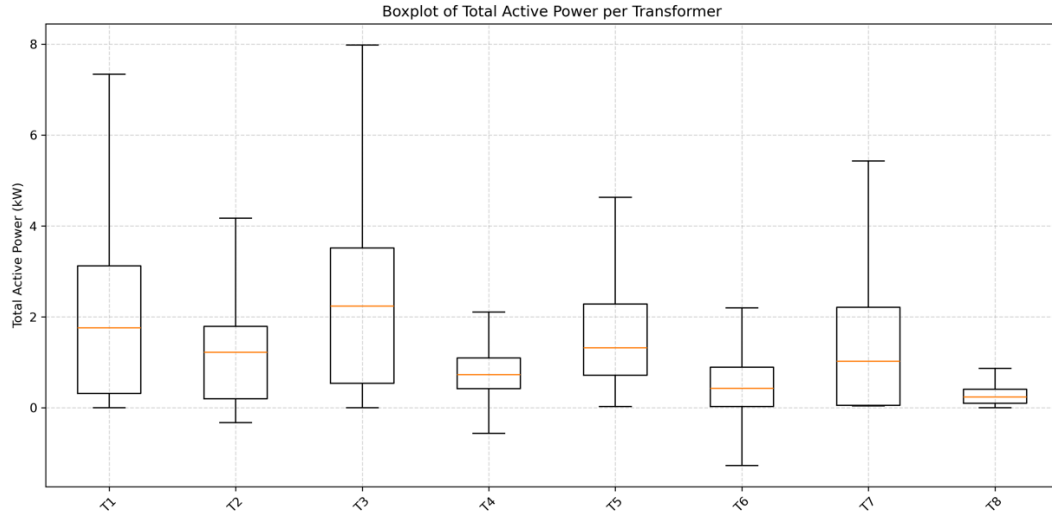


Figure 12: Total active power (kW) per transformer

After obtaining a better understanding of the dataset, the hourly averages for each transformer by grouping the data by hour of the day were computed. The profiles showcased new insights on consumption.

4.2.2 Hourly demand Curves and Duck curve visualization

After cleaning the dataset, hourly load profiles for each of the eight transformers were generated using Python as shown in Figure 13. The average active power for each hour of

the day is calculated by aggregating all consumption data across the entire period for each transformer.

Hourly consumption patterns were also extracted and created for individual households. The household's individual data was processed in the same way as the transformer data using timestamp conversion and summing total active power by hour. Comparison between (transformer-based) and bottom-up (household-based) demand was conducted to check for errors in the datasets. The household-level aggregation matched the transformation data curve. However, individual load curves displayed variances in loads. Some individuals had a consumption of 0, which shows that they are either not connected to the grid or not residing in their Bechaaleh home. Other individuals showcased atypical consumption patterns (loads that do not follow the duck curve). These patterns could also be because that some households rarely spend time at their Bechaaleh home or turn off electricity at night, which creates unusual consumption patterns. This reaffirms the importance of studying the system as a whole.

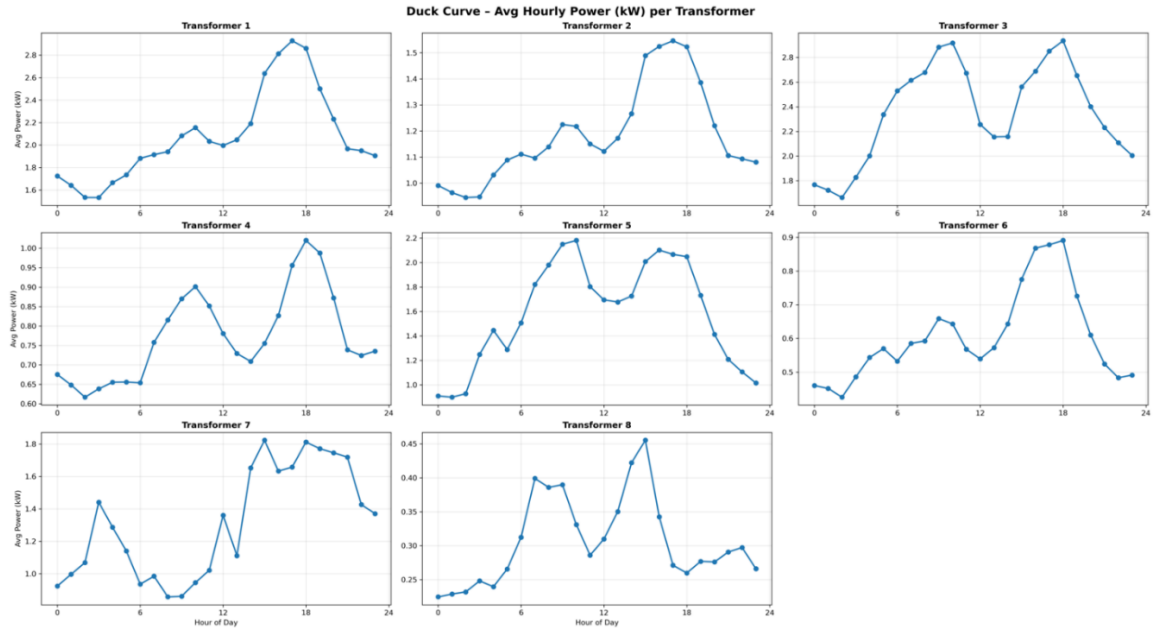


Figure 13: Average hourly Power (kW) per transformer in Bechaaleh

Most of the transformers show two characteristic demand peaks, which are a modest one in the early morning and a sharper one in the evening. The patterns observed align with

typical household usage patterns, as consumption increases in the morning when people wake up, and after sunset when lighting, heating, and cooking are more prevalent. Transformers 3 and 5 are the ones that exhibit the most pronounced duck curves, as they showcase a noticeable decrease in average consumption during midday hours, around 11 AM to 2 PM. Moreover, transformers with fewer users such as transformers 7 and 8 exhibit more irregular shapes and have lower overall consumption. This is consistent with expected results, as these transformers have limited households attached to them. The sum of consumption per hour is also analyzed, and the resulting plots are consistent with the average power analysis.

Since the transformer curves reflect the energy consumption of several households, to obtain a better understanding of the data, the whole system needs to be analyzed. Around 150 individual curves were plotted (Figure 14).

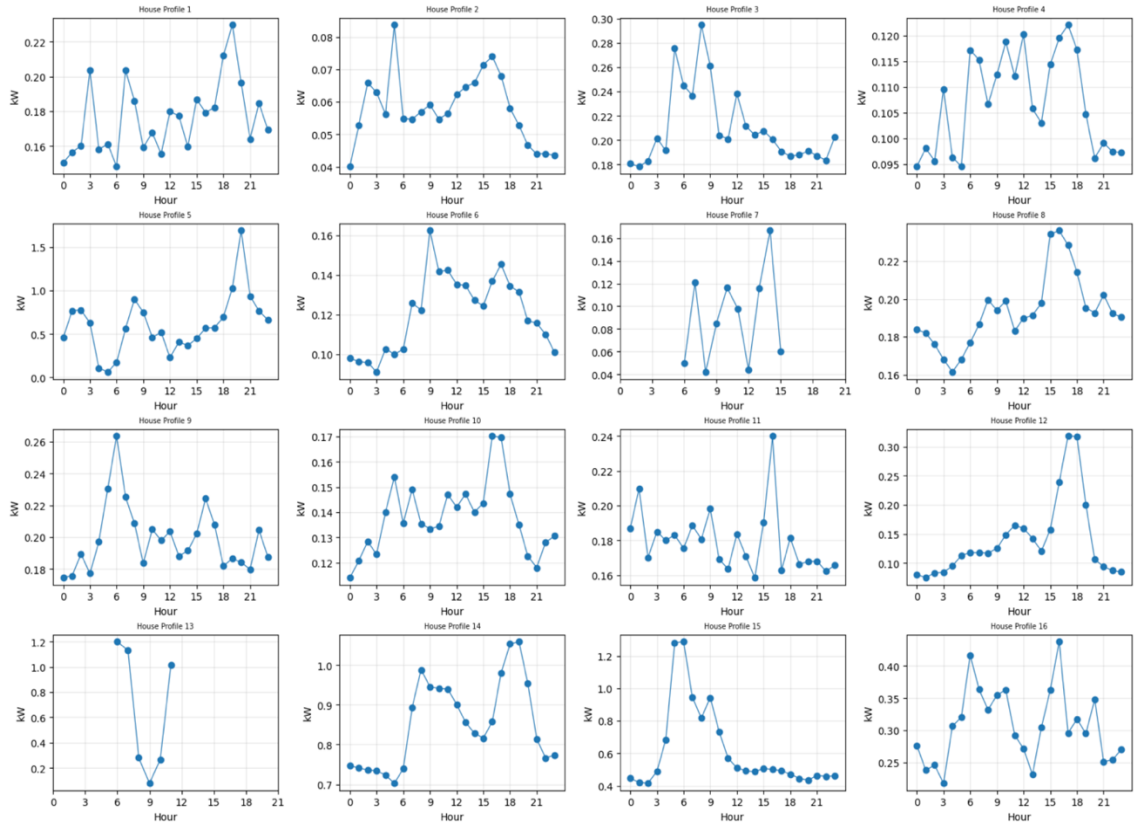


Figure 14: Extract of the output showing some housing profiles (kW) of Bechaaleh

To identify which scenario of the previously proposed model best fits Bechaaleh's electricity demand, two approaches were applied. The first approach is the dominant trend approach, which consists of identifying the most recurring trend by focusing on the shape and timing of the consumption peaks. A representative household trend will be compared to the 12 scenarios of our model. Moreover, the second approach consists of creating an average of all individual household profiles, which shall also be compared to the 12 scenarios of the model. All hourly profiles were plotted in Figure 15 together to attempt to identify a global trend in consumption. At first glance, while plotting all hourly profiles, some outliers are noticed. Indeed, around 4 households have hourly instantaneous power over the value of 1 kW. Outliers are excluded to obtain a global trend after conducting an initial filtering to exclude unnecessary noise. After conducting the initial filtering with profiles with hourly instantaneous power up to 1 kW, realistic consumption patterns are observed, and households with higher demand profiles are included. Although the overall median remains relatively flat, a slight binodal demand pattern can be observed.

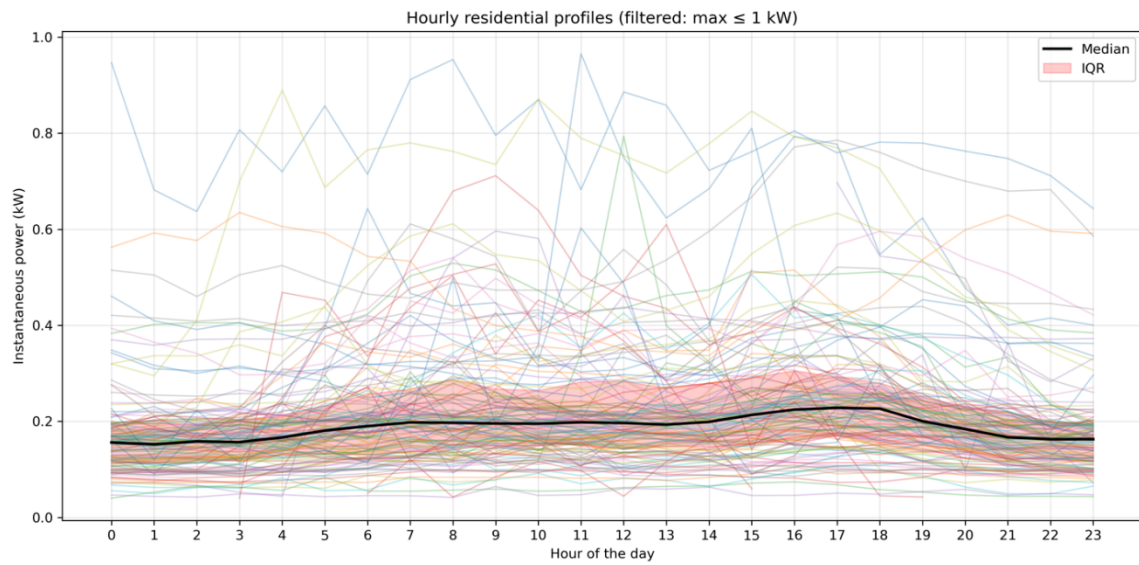


Figure 15: Demand trend for hourly daily profile (kW) (Filtered with values ≤ 1 kW)

After noticing that most profiles have values falling between 0.05 and 0.35, a separate plot shown in Figure 16 was created that considers profiles with instantaneous power between 0.05 kW and 0.35 kW. The median curve shows a moderate increase during the morning

hours, before reaching a midday dip. A visible evening peak can be noticed between 17:00 and 20:00. The interquartile range displays the variability amongst households, with stronger variability in the afternoon. The black median curve characterizes the global consumption trend that will be studied.

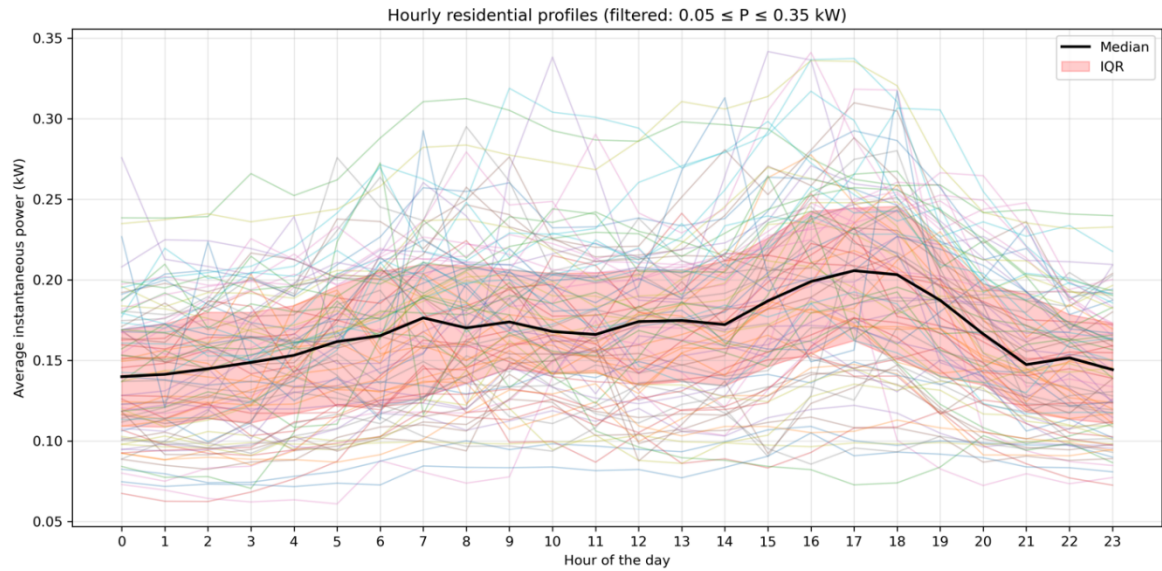


Figure 16: Demand trend for hourly daily profile (Filtered with values between 0.05 and 0.35 kW)

Additionally, an average hourly load for a Bechaaleh household is created and shown in Figure 17, which consists of the mean hourly instantaneous power in kW across all Bechaaleh profiles. The results across both approaches are consistent, as the village-wide mean average curve showcases a comparable load curve. Both approaches, one depending on a dominant trend and the other on the mean average load trend, showcased a binodal pattern, which reaffirms that the trends observed are reliable.

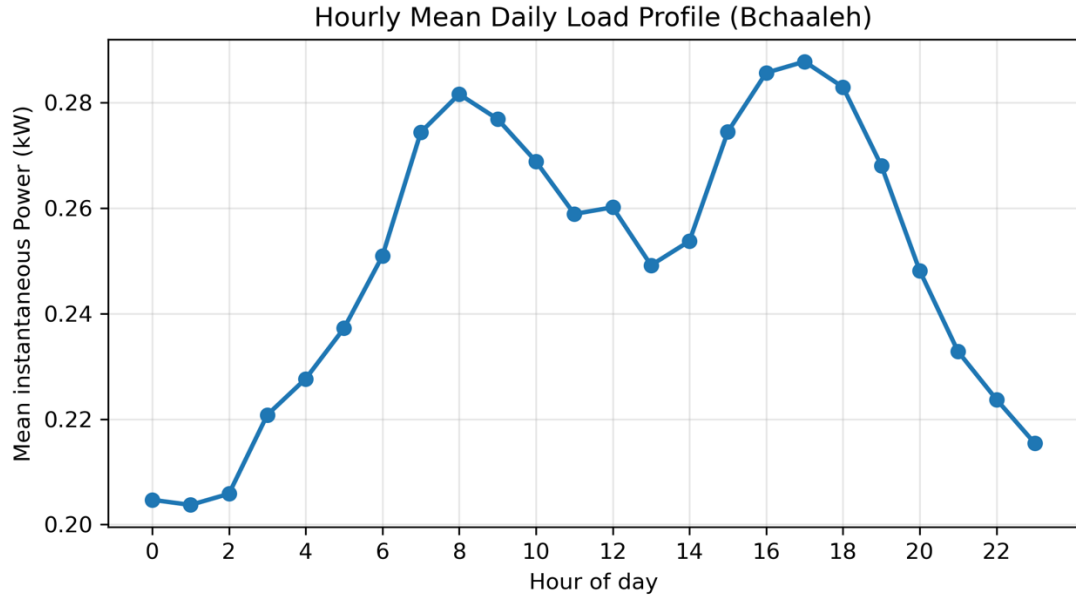


Figure 17: Average hourly load for a Bechaaleh household

The monthly consumption in kWh is also analyzed. Since the dataset does not cover the entire year, a complete interpretation of monthly trends could not be conducted. However, the consumption patterns show that some households are away during winter as many transformers display lower consumption between December and February (Figure 18). These patterns are consistent with villages where residents relocate to the city during the year and spend summers at their respective villages.



Figure 18: Monthly consumption per transformer

An aggregated village-wide monthly energy consumption is also created and plotted in Figure 19.

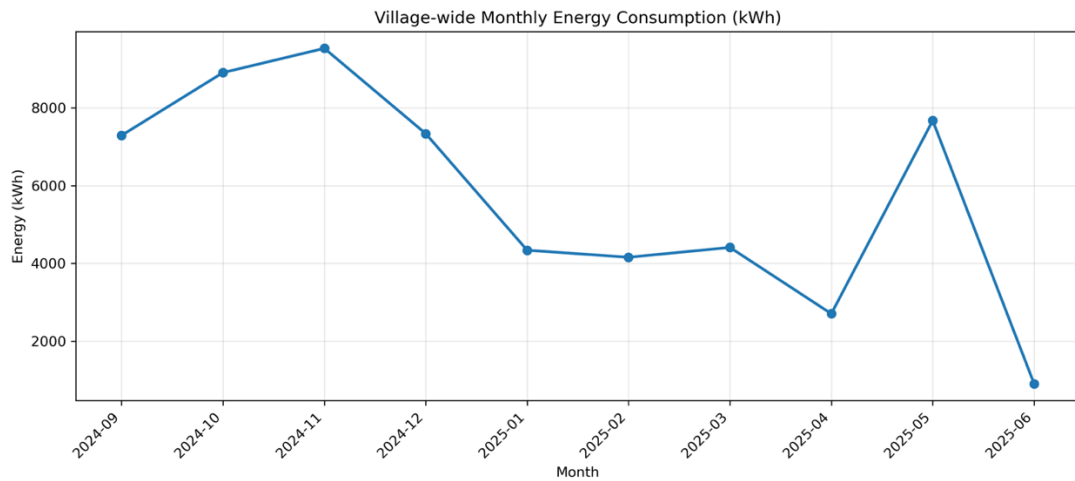


Figure 19: Village wide monthly consumption

4.3 Model Validation

4.3.1 Daily Load curves validation

4.3.1.1 *Compared to National Household values*

To analyze model accuracy, the average daily load curve from the simulation is compared with Bechaaleh's data. Both curves exhibit the characteristic two-peak shape, but differences in peak timing, shape, and intra-day fluctuations can be assessed. In the real data, the evening peak tends to occur around sunset hours (around 17:00-20:00) and is relatively sharp, reflecting synchronized household activities after work such as lighting, cooking, and heating. Most transformers in the village indeed show a pronounced rise in the early evening, preceded by a smaller uplift in the early morning when residents start their day. The bottom-up model reproduces a comparable pattern: its composite daily profile peaks in the early night, around 7-8 PM, with a secondary rise at dawn as lights and appliances turn on. The peak timings in the model align to within an hour or two of the observed peaks, which is reasonable given potential differences in occupant schedules. Both profiles also show a clear mid-day depression in demand (period of low consumption), consistent with many occupants being away or less active at mid-day.

In terms of shape and fluctuations, the model's daily curve is somewhat smoother than the real-world curve for individual households, as expected. The empirical average profile sits amid a broad band of individual household curves (see Figure 16 above), with an afternoon interquartile range in the order of 0.1-0.25 kW, indicating some homes spike usage while others stay low. The simulation, by aggregating many household types, naturally produces a smoother mean with less random noise. Importantly, the peak-to-trough amplitude of the daily cycle is captured well: both the model and measurements show roughly a 2 to 3 times increase from mid-day low to evening high. One discrepancy is in the late-night hours: the model maintains a relatively elevated plateau of demand until around midnight (assuming many appliances and lighting remain in use), whereas the Bechaaleh data show a slight decline after 20:00 as some households taper off usage. Despite such minor differences, the day load shape generated by the model mirrors the observed pattern in all essential features. The morning increase, mid-day trough, and

evening peak are present in both, indicating that the time-of-use behavior coded into the simulation (occupancy patterns, appliance schedules) is realistic for the village.

4.3.1.2 Compared to Zone 4 average:

A binodal demand can be observed in zone 4, similarly to the average Lebanese consumption. This pattern is similar to the dominant trend in Bechaaleh's data. Indeed, Figure 16 shows a median profile that displays a dual-peak shape with reduced consumption during the middle of the day. In both the model and Bechaaleh, the timing of the peaks is consistent with a rise in consumption in the evening. As with national average comparison, in the model, the amplitude of the peak is higher than that of Bechaaleh. This is also due to the assumptions of continuous occupancy. In reality, households showcase a greater variability and a more gradual reduction of consumption at night.

4.3.2 Monthly Energy Consumption (kWh)

Lastly, a comparison between monthly aggregated consumption of the model and the Bechaaleh data is conducted to check seasonality and total energy accuracy. Here data limitations were accounted for. The available smart meter records span from late 2024 to mid-2025, leaving some months of the year unobserved. Consequently, a full year comparison is not possible. However, clear trends emerge in the partial data. The real measurements indicate significantly lower energy use during winter months (approximately December-February) and higher consumption moving into late spring and summer. Many households in Bechaaleh appear to be vacant or use minimal power in winter, a pattern consistent with families relocating to the city during the cold season and returning in summer. This could be one the reason why the peak consumption of the model is higher than that of Bechaaleh.

4.3.2.1 Compared to the Lebanese household

When quantifying monthly kWh, the model's predictions are compatible with the Bechaaleh data for the available months. In January and February, both model and data show depressed usage. The model slightly overestimates these winter months' consumption, likely because it assumes a baseline occupancy/usage in every home, whereas in reality, some homes in Bechaaleh drew virtually zero power in that period as they are unoccupied. By spring, the gap between modeled and observed monthly totals narrows as more households in Bechaaleh become active and the model's inclusion of cooling loads begins to kick in. The model's predicted summer peak (about July–August) provides a reasonable explanation for the rising trend observed in the partial dataset, although there is insufficient recorded data for all summer months for validation. Overall, the seasonal consumption pattern is well represented by the model since it captures the winter decrease and summer surge.

4.3.2.2 Compared to zone 4

In zone 4, summer loads are low given the lack of air conditioning at high altitude. Winter showcases high demand due to lighting and heating. The Bechaaleh data has confirmed this finding. Indeed, the dataset shows lower winter consumption mainly due to absent households. However, the model still reflects peaks. Discrepancies between the model and zone 4 come from the fact that the model does not capture vacancy behavior as many Bechaaleh residents may be away during winter.

4.3.3 Model validity and limitations

Based on the above comparisons, the bottom-up household demand model is partially validated by the Bechaaleh data. The Bechaaleh data reproduces a pronounced duck curve shape. These patterns match the model's structure in terms of appliance usage cycles and occupancy assumptions. The average of all simulated scenarios provides a close fit to the average daily load in Bechaaleh. Compared to zone 4, the model's load profile aligns well

with the Bechaaleh data in terms of timing and shape. This highlights that the assumptions taken are valid for the high mountain context. The model, however, still needs refinement for this specific case by integrating household presence probability.

The model still exhibits some weaknesses and limitations, which suggest that it could benefit from further tuning. One limitation is the data itself: the real smart-meter dataset is incomplete (missing certain months and excluding 2025's latter half), which makes it challenging to fully validate annual energy predictions. Therefore, patterns are the main concern of the comparisons rather than exact quantities. Additionally, there are structural assumptions in the model that simplify reality. For instance, the model assumes every household has a consistent occupancy pattern and a baseline usage. The model does not explicitly simulate houses that go entirely dark for weeks or months. In the Bechaaleh data, a subset of homes had essentially zero consumption for extended periods (owners not present or disconnecting power), a behavior the current model does not capture. This led to small discrepancies, such as the model over-predicting winter demand when many real homes were empty. Similarly, the model maintains a steady late-night usage in all scenarios, whereas real households show a slight decline as night progresses, showing that sleep or shut-off behaviors could be more pronounced than modeled.

In sum, the model captures the shape of demand curves and seasonal effects correctly but exhibits some deviation in magnitude and in handling certain cases. For example, a probability of seasonal vacancy could improve the fit.

5 Conclusion and perspectives

The aim of the thesis is to create and validate a bottom-up model for residential electricity demand of the Lebanese population. The model uses appliance schedules and Lebanon's four climate zones to create 12 household scenarios. These scenarios are characterized by 3 dwelling sizes and four climate zones assumptions. An average Lebanese household profile was created by combining different weights of the 12 scenarios. The Bechaaleh smart meter dataset daily load curve as well as the yearly consumption curve were tested

against the model's Lebanese average load curves as well as a zone 4's load curves. Bechaaleh's dataset was therefore used to validate the model and its assumptions.

5.1 What the model accomplishes

Using Python, the model includes a code that takes as input schedules appliances such as lighting, refrigeration, TV, washing, electric water heating, AC, and "other" spikes. These schedules follow rules that vary by hour, season, and usage intensity. The rules are created based on the characteristics of Lebanon's four climate zones. These zones differ in consumption patterns such as AC usage and heating. The simulation runs at an hourly resolution over the entire year (8760 hours). The model also generates daily and seasonal curves. The assumptions used in the model as well as its outputs are documented in the model and figures described above. These assumptions serve as a reusable framework for scenario analysis. Twelve scenarios combining three dwelling sizes areas (80, 120, 200 m²) with four climate zones are created. By combining the different scenarios, village-level average load curves are created. The results show consistent patterns of residential demand with a peak in demand in the morning and in the evening.

The model also handles seasonality. Zone definitions explicitly include the intensity of cooling in summer, as well as lighting water heating. Electric space heating is included, since the model assumes that in mountainous zones, households rely on diesel or wood heating. Therefore, our assumptions have led to inland areas exhibiting the strongest summer peaks and zone 4 modest summer loads.

5.2 Comparison between Bechaaleh's data and the model

Two complementary validation approaches were applied for the Bechaaleh smart-meter data. The first approach is the dominant-trend approach using the median of filtered hourly profiles and the second approach is a daily mean load profile. Both profiles highlight a

bimodal daily pattern with a clear evening peak and a smaller morning peak. The profiles exhibited a duck curve pattern of residential demand.

Incomplete smart-meter coverage with missing monthly prevented the conduction of a complete yearly consumption analysis. However, lower consumption in winter can be noticed. Slight differences in magnitude can be observed since the model assumes occupancy of household. Given Bechaaleh's altitude, it is better suited to compare it with zone 4 of our model. Zone 4 analysis emphasizes cool summers with limited AC use and modest seasonal differences in consumption. The village data also showcases this pattern.

5.3 Key insights for the grid

a. Load shape and flexibility:

The evening peak dominates village demand. Separating inflexible from flexible end-uses suggests is practical for demand response. For example, water heating and laundry can be shifted to reduce the peak without compromising essential services. This is highly actionable for microgrid scheduling and storage sizing.

b. Zone heterogeneity matters:

Zone 3 households need summer AC capacity. Zone-4 households require reliable evening energy for lighting and hot water rather than large summer cooling capacity. Therefore, the system design should be geographically differentiated for each region.

c. Seasonal patterns:

The three-phase yearly pattern: stable shoulder seasons, a summer rise (or muted rise in Zone 4), then decline, affects storage and PV. This highlights the importance of validating both daily and seasonal scales, not just annual kWh.

d. Validation based on evidence:

Using both a median-trend and a mean-trend from Bechaaleh avoids validating our model with a single statistic method. Moreover, this improves confidence since it ensures that our model does not include either extreme profiles or averaging alone.

5.4 Limitations of the study

Several simplifying choices temper the interpretation:

a. Occupancy and vacancy:

The model assumes continuous baseline presence and only general work-hour absence, whereas Bechaaleh shows pronounced seasonal vacancy, especially in winter, which leads to lower measured kWh than predicted. Incorporating household-level presence probabilities such as day-type, season, and residency status would improve amplitude fidelity.

b. Appliance usage granularity:

Schedules are rule-based (not stochastic per-appliance processes) and sometimes too rigid. Indeed, households do not use appliances in the exact timeslots defined for the model. This smooths the variability seen across households and transformers. Incorporating probabilities of use for appliances would better reproduce dispersion and occasional spikes.

c. Data completeness:

The smart-meter series omits some months, constraining robust monthly validation and year-over-year checks. Continued data collection would enable stronger calibration and counterfactual experiments (e.g., DR trials, PV penetration curves).

d. Scope of electric loads:

Electric space heating is excluded by design because it is uncommon outside coastal apartments. If the electrification of heat in the mountain accelerates, the model would need to be extended to capture winter peaks from heat pumps.

e. Generalization limited to zone 4:

The model is validated against zone 4. While Zones 1 to 3 are modelled, they are not validated separately, given the lack of data.

5.5 Perspectives and future work

- **Calibrate presence and diversity:** Integrating household-level “presence processes” (weekday/weekend, school breaks, summer return to village) and seasonal occupancy priors for second homes could enrich the study. This will decrease winter loads where vacancy is likely. It can be conducted through occupancy surveys.
- **Incorporate probability of appliance usage:** Replacing deterministic time schedules with probabilistic start-time distributions and cycle durations per device could be included in the study. For water heaters and laundry, randomization around typical hours could also be incorporated.
- **Seasonal/temperature calibration:** Using measured temperature and daylight for the study period and calibrating the AC threshold by season to add more precision to the model.
- **Flexible-load experiments:** Testing demand response could be incorporated to quantify peak reduction potential and decrease reliance on conventional energy sources.
- **Expand the validation set:** Adding more villages across zones 1 to 3 for validation could expose some biases and allow the model to become more generalizable.
- **Conducting the supply analysis:** Couple the demand model to microgrid supply (PV, storage, diesel backup) to design a DR program that flattens the duck curve. Supply analysis constitutes an essential step in the study to match supply and demand. However, given Lebanon’s current crisis as well as the unreliability of EDL as a viable source of electricity supply, obtaining accurate data becomes challenging.
- **Open, replicable code:** Packaging the code with default parameters and plots that fit each scenario, or creating a code that automatically generates charts regardless of the dataset type, such as seasonal 2×2 daily curves, yearly kWh, and flexible-load partitions, could be an interesting addition to the study.

5.6 Final assessment

To conclude, the model reproduces the shape and timing of Bechaaleh's daily demand convincingly, including the evening peak and mid-day trough characteristic of residential communities. The model includes some bias that increases winter consumption due to simplified occupancy. This is addressable with presence modeling and calibration to observed seasonal vacancy. By incorporating these additions, the model would become more fit to be used as a basis for creating a dynamic tariff. Indeed, it would highlight that flexible load might be available for peak shaving and how zone-specific climates shape seasonal needs. With the enhancements outlined above, the validation could become stronger and the model more robust.

6 Code appendix

6.1 Demand modelling code

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
from math import sin, pi
import random

# --- Constants ---
DAYS_IN_YEAR = 365
HOURS_IN_DAY = 24
TOTAL_HOURS = DAYS_IN_YEAR * HOURS_IN_DAY
MONTH_DAYS = [31,28,31,30,31,30,31,31,30,31,30,31]

climate_zones = {
    1: {"name": "Coastal", "summer_daily_high": 32, "winter_daily_high": 18},
    2: {"name": "MidMountain", "summer_daily_high": 30, "winter_daily_high": 12},
    3: {"name": "Inland", "summer_daily_high": 36, "winter_daily_high": 10},
    4: {"name": "HighMountain", "summer_daily_high": 25, "winter_daily_high": 5}
}

appliance_specs = {
    "Lighting": 0.01,
    "Refrigerator": 0.15,
    "TV": 0.10,
    "WashingMachine": 0.50,
    "WaterHeater": 1.50,
    "AC": 1.00,
    "Others": 1.00
}

# --- Helper Functions ---
def get_month(day_of_year):
    for i, days in enumerate(MONTH_DAYS):
        if day_of_year <= days:
            return i + 1
    day_of_year -= days
```

```

return 12

def lighting_schedule(hour, day, area):
    if hour >= 18 or hour < 6:
        month = get_month(day)
        seasonal = 1.2 if month in [12,1,2] else 1.0
        behavior = 1.1 if area > 150 else 1.0
        return appliance_specs["Lighting"] * area * seasonal * behavior
    return appliance_specs["Lighting"] * area * 0.01

def ac_schedule(hour, day, zone, occupants_home):
    z = climate_zones[zone]
    summer = 152 <= day <= 273
    high = z["summer_daily_high"] if summer else z["winter_daily_high"]
    temp = high if not summer else high - 10 + 10 * sin(pi * (hour - 6) / 12)**2
    return appliance_specs["AC"] if temp > 21 and occupants_home else 0

def water_heater_schedule(hour, day, usage):
    month = get_month(day)
    if usage == "high":
        return appliance_specs["WaterHeater"] if 6 <= hour < 8 or 18 <= hour < 20 else 0
    if usage == "medium":
        return appliance_specs["WaterHeater"] if 6 <= hour < 8 or (18 <= hour < 20 and
month in [11,12,1,2]) else 0
    return appliance_specs["WaterHeater"] if 6 <= hour < 8 else 0

def tv_schedule(hour, dow, usage):
    if usage == "high" and 8 <= hour < 24:
        return appliance_specs["TV"]
    if usage == "medium" and 18 <= hour < 23:
        return appliance_specs["TV"]
    if usage == "low" and 19 <= hour < 22:
        return appliance_specs["TV"]
    return 0

def laundry_schedule(day, hour, usage):
    per_week = {"low": 2, "medium": 4, "high": 7}[usage]
    prob_hour = per_week / (7 * 12)
    if 8 <= hour < 20:
        prob = prob_hour * 1.5 if day % 7 in [5,6] else prob_hour

```

```

        return appliance_specs["WashingMachine"] if random.random() < prob else 0
    return 0

def others_schedule():
    return 0.02 + (appliance_specs["Others"] if random.random() < 0.01 else 0)

# --- Household Profile Generator ---
def generate_household_profile(zone, area):
    usage = "low" if area <= 80 else "medium" if area <= 120 else "high"
    ac_units = 0 if zone == 4 and usage == "low" else (1 if area <= 80 else 2 if area <= 120
else 3)
    fridge_count = 1 if area <= 120 else 2
    has_water_heater = not (zone == 4 and random.random() < 0.5)

    profile = np.zeros(TOTAL_HOURS)
    for day in range(DAYS_IN_YEAR):
        dow = day % 7
        for hour in range(HOURS_IN_DAY):
            idx = day * HOURS_IN_DAY + hour
            occupants_home = not (dow < 5 and 9 <= hour < 17)
            load = 0
            load += lighting_schedule(hour, day, area)
            load += fridge_count * appliance_specs["Refrigerator"]
            load += tv_schedule(hour, dow, usage)
            load += ac_units * ac_schedule(hour, day, zone, occupants_home)
            if has_water_heater:
                load += water_heater_schedule(hour, day, usage)
            load += laundry_schedule(day, hour, usage)
            load += others_schedule()
            profile[idx] = round(load, 3)
    return profile.tolist()

```

6.2 Bechaaleh demand code - Transformer analysis figure 13

```

import pandas as pd
import matplotlib.pyplot as plt
import os

# === Paths and transformer list ===

```

```

before_path = "Before_2025"
after_path = "After_2025"
transformer_ids = ["1", "2", "3", "4", "5", "6", "7", "8"]

# === Dictionary to store hourly profiles
hourly_profiles = {}

# === Process each transformer individually
for trans_id in transformer_ids:
    # Fix file naming for transformer 3
    suffix_before = f'{trans_id}-20230101-20241231.csv' if trans_id != "3" else
f'{trans_id} -20230101-20241231.csv'
    suffix_after = f'{trans_id}-20250101-20250605.csv' if trans_id != "3" else
f'{trans_id} -20250101-20250605.csv'

    before_file = os.path.join(before_path, f'E-Trans {suffix_before}')
    after_file = os.path.join(after_path, f'E-Trans {suffix_after}')

    # Read and merge
    df_before = pd.read_csv(before_file)
    df_after = pd.read_csv(after_file)
    df = pd.concat([df_before, df_after], ignore_index=True)

    # Cleanup
    df["EntryDT"] = pd.to_datetime(df["EntryDT"], errors="coerce")
    df = df.dropna(subset=["ActivePowerP1", "ActivePowerP2", "ActivePowerP3"])
    df = df.sort_values("EntryDT")

    # Compute total instantaneous 3-phase power
    df["TotalActivePower_kW"] = df["ActivePowerP1"] + df["ActivePowerP2"] +
df["ActivePowerP3"]

    # Round timestamp to hour
    df["HourTimestamp"] = df["EntryDT"].dt.floor("h")

    # Sum or average all points within the same hour
    hourly
    df.groupby("HourTimestamp")["TotalActivePower_kW"].sum().reset_index()

# Extract hour of day

```

```

hourly["HourOfDay"] = hourly["HourTimestamp"].dt.hour

profile=
hourly.groupby("HourOfDay")["TotalActivePower_kW"].mean().reset_index()
    hourly_profiles[f'Transformer {trans_id}'] = profile

# === Plot all hourly profiles (3x3 layout) ===
fig, axes = plt.subplots(nrows=3, ncols=3, figsize=(22, 12), constrained_layout=True)
axes = axes.flatten()

for i, (name, df_profile) in enumerate(hourly_profiles.items()):
    ax = axes[i]
    ax.plot(df_profile["HourOfDay"], df_profile["TotalActivePower_kW"], marker='o',
linewidth=2)
    ax.set_title(name, fontsize=12, fontweight='bold')
    ax.set_xticks(range(0, 25, 6))
    ax.set_xlabel("Hour of Day")
    ax.set_ylabel("Power (kW)")
    ax.grid(alpha=0.3)

# Hide the unused 9th subplot
for j in range(len(hourly_profiles), len(axes)):
    fig.delaxes(axes[j])

fig.suptitle("Duck Curve – Sum of Hourly Power (kW) per Transformer", fontsize=16,
fontweight='bold')
fig.savefig("duck_curve_per_transformer.png", dpi=300)
plt.show()

```

6.3 Bechaaleh demand code - Individual household analysis figure 16

```

import os, re
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path

before_dir = Path("Before_2025")
after_dir = Path("After_2025")

```



```

def list_individual(folder: Path):
    return [p for p in folder.iterdir()
            if p.name.startswith("M-") and p.suffix == ".csv"]

def root_name(fname: str) -> str:
    return re.sub(r"-d{6,8}.*\.csv$", "", fname)

def clean(df: pd.DataFrame) -> pd.DataFrame:
    df = df.dropna(axis=1, how="all")
    df.columns = [c.strip() for c in df.columns]
    return df

def find_entrydt(df: pd.DataFrame) -> str | None:
    for c in df.columns:
        if c.replace(" ", "").lower().startswith("entrydt"):
            return c
    return None

P_COLS = ["ActivePowerP1", "ActivePowerP2", "ActivePowerP3"]

pairs = {}
for f in list_individual(before_dir):
    pairs.setdefault(root_name(f.name), {})[before] = f
for f in list_individual(after_dir):
    pairs.setdefault(root_name(f.name), {})[after] = f

def hourly_profile(paths: dict) -> pd.Series | None:
    if {"before", "after"} - paths.keys():
        return None

    df = pd.concat(
        [clean(pd.read_csv(paths[before])),
         clean(pd.read_csv(paths[after]))],
        ignore_index=True, join="inner"
    )

    ts = find_entrydt(df)
    if ts is None or not set(P_COLS).issubset(df.columns):
        return None

```

```

df[ts] = pd.to_datetime(df[ts], errors="coerce")
df = df.sort_values(ts).dropna(subset=[ts]).drop_duplicates(ts)
df["Ptot"] = df[P_COLS].sum(axis=1, skipna=True)
df = df[df["Ptot"] > 0]

if df.empty:
    return None
if df["Ptot"].median() > 1000:
    df["Ptot"] /= 1000

df["Hour"] = df[ts].dt.hour
hourly_avg = df.groupby("Hour")["Ptot"].mean()

# FILTER: keep only if ALL values between 0.05 and 0.35 kW
if (hourly_avg < 0.05).any() or (hourly_avg > 0.35).any():
    return None

return hourly_avg

all_profiles = []
for meter, pathdict in pairs.items():
    prof = hourly_profile(pathdict)
    if prof is not None and not prof.empty:
        all_profiles.append(prof)

if all_profiles:
    df_profiles = pd.concat(all_profiles, axis=1)
    df_profiles.columns = [f"Household_{i+1}" for i in range(len(df_profiles.columns))]

plt.figure(figsize=(12, 6))
for col in df_profiles.columns:
    plt.plot(df_profiles.index, df_profiles[col], alpha=0.3, linewidth=1)

median_curve = df_profiles.median(axis=1)
q1 = df_profiles.quantile(0.25, axis=1)
q3 = df_profiles.quantile(0.75, axis=1)
plt.plot(df_profiles.index, median_curve, color="black", linewidth=2, label="Median")
plt.fill_between(df_profiles.index, q1, q3, color="red", alpha=0.2, label="IQR")

```

```
plt.title("Hourly residential profiles (filtered:  $0.05 \leq P \leq 0.35$  kW)")
plt.xlabel("Hour of the day")
plt.ylabel("Instantaneous power (kW)")
plt.xticks(range(0, 24, 1))
plt.grid(alpha=0.3)
plt.legend()
plt.tight_layout()
fig.savefig("average_0..png", dpi=300)
plt.show()
else:
    print("No profile for 0.05 - 0.35 kW.")
```

7 References

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